A Finite-Shadow Grammar for Finite Centered Incidence Geometry

Oleksiy Babanskyy

Abstract

This paper records a bounded finite-shadow grammar for finite centered incidence geometry. A row instance is admitted as a tuple $\mathcal{S} = (N, X, I, \rho, \Phi)$, where N is the ambient symmetric degree and $X \subseteq S_N$; the labels X01–X26 name source families rather than individual subsets. The five-slot core is retained, while a normative typed source lock fixes provenance, action conventions, arithmetic and quotient data when present, verification, ownership, and import boundaries. The payoff is not a universal theory: it is a controlled interface that makes standard-block, modular, source-partition, source-layer, and fingerprint tests visible across Type-A, signed Type-B/C, dihedral, G_2, projective F_4, H_3, H_4, D_5, E_6, E_7, and E_8 finite rows, together with centered-operator exhibits. The standard-block dictionary and the Smith-form modular rank bridge are elementary, but useful here because they are applied to source-locked rows: for example, the modular row records a carrier-coprime torsion collapse, with $2 \nmid 7$, where $I_7(1, 1)$ has rational augmentation rank 3 but rank 0 over $\mathbb{F}_2$. The worked rows show that Φ is not determined by coarse histograms, rational rank alone, characteristic-zero data alone, uncolored ambient conjugacy alone, or the finite set alone. The paper also records the limits: Φ is not a complete invariant, not a tiling oracle, and not a substitute for the separate theorem and certificate routes.

Keywords: finite centered incidence geometry; finite shadows; symmetric groups; Specht fingerprints; centered operators; modular centered channels; signed permutations; dihedral groups; source partitions; G2 root shadows; F4 root shadows; H3 root shadows; H4 root shadows; D5 root shadows; E6 root shadows; E7 root shadows; E8 root shadows.

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1 Introduction and Boundary

Finite centered incidence geometry studies finite incidence-defined rows together with the representation or centered-operator channels that make those rows comparable. The purpose of this paper is to record a bounded grammar for such rows. It is not a final theory of every FCIG branch, and it does not turn the fingerprint channel into a complete invariant.

The paper is built after public source rows. The Type-A standard-space paper supplies the first mature theorem row[Bab26f]. The S_4 finite witness paper supplies a small public finite witness for the structured-subset layer[Bab26i]. The Type-A poset-cone and extension-tiler paper supplies the public chamber/tiler context for rows X01–X02[Bab26l]. The Latin-square and Sudoku operator papers supply public centered-operator sibling context for rows X05–X06[Bab26b, Bab26j]. The present paper does not reopen those proofs; it uses them as source rows or related-work rows in a broader grammar.

The organizing principle is that a row is admitted only after it has a finite ambient set, a stated incidence constraint, a representation or operator channel, and a recorded fingerprint package. Rows without such a finite shadow remain outside the core grammar.

The modular centered row is included as a compact channel exhibit rather than as a new theorem route. It promotes the reserved X16 row: the same integral centered operator can have one rational augmentation rank and a different rank after reduction modulo a prime. The key source-locked witness is $I_7(1,1)$ in characteristic 2, where 2 does not divide 7, the lattice-qualified Smith form is $\mathrm{SNF}^L = [4,4,8]$, and the augmentation rank drops from 3 over $\mathbb{Q}$ to 0 over $\mathbb{F}_2$. The point is that a finite-shadow row may need characteristic, augmentation channel, reduced quotient, and lattice-qualified Smith data, not rational rank alone. The new dihedral row in this revision is included for that reason: it is a finite vertex action $I_2(m) \leq S_m$, with explicit incidence rows and replayed standard-block fingerprints. Its mathematics is elementary cyclic and dihedral linear algebra[BB05]; its role here is to test whether the grammar still makes useful distinctions in a rank-two reflection shadow. The length-colored G_2 row is included with the same restraint. It is not a general Weyl theory; it is a finite root-shadow exhibit showing that the grammar sometimes has to remember source channels, here the short/long partition, rather than only the uncolored ambient permutation set.

The projective F_4 row adds a second, exceptional source-aware test. It uses the 24-point projective root shadow and the matroid automorphism source of Fried–Gerek–Gordon–Perunicic[FGGP07]. The row is included only to compare two source layers on the same finite set: the geometric projective Weyl layer preserves the long/short split and leaves one centered length residual, while the source matroid automorphism layer adds a non-geometric length-swap coset and cancels that residual. This is not a theorem about all F_4 rows, Weyl groups, or the 24-cell.

The projective H_3 row is included in the same compact spirit, but with a different channel split. On the same 15 projective root pairs, the flat-size/matroid channel has automorphism group 120, while the Gram-angle geometric channel has automorphism group 60. The ordinary standard block is blind to this boundary, and the first higher ordinary Specht separator is $(13,2)$. The row is used only as a finite source-channel exhibit, not as a theorem about all H_3 shadows or non-crystallographic Coxeter systems.

The projective H_4 row is included only after the selected-atlas source package closes its finite channel checks. It uses the 60 antipodal pairs in the H_4 root system, viewed as a projective 600-cell root shadow, and compares two pair-color channels: rank-two flat size and squared Gram value. The flat-size channel has full automorphism group 14400, while the Gram-square channel has full automorphism group 7200. This is the source-aware lesson used here; it is not a paper about H_4, the 600-cell, or root-system matroid automorphism theory. The projective D_5 row is included as a control on the pair-channel boundary. Unlike G_2 or F_4, it has no short/long length split. The point is that pair-level Gram/flat-size data still has too many automorphisms: $C_2^{10} : S_5$ of order 122880. The 40 projective A_2 3-flat incidences cut this back exactly to the projective Weyl action of order 1920. This is not a standalone D_5 theorem and not a transfer to E_6, E_7, E_8.

The projective E_6 row is included as a second simply laced control, but with the opposite lesson. The 36 projective root pairs have pair, flat-size, and circuit channels whose full automorphism groups all agree with the 51840-element projective source action. The useful grammar datum is therefore not an enlarged automorphism group, but the source-aware decomposition $\mathbb{Q}^{\Pi_{36}} = \mathbb{Q}\mathbf{1} \oplus U_{20} \oplus U_{15}$: two parabolic rows can have the same size and total standard rank while splitting differently across U_{20} and U_{15}. This is not a standalone E_6 theorem and not a Schlaefli, double-six, 27-line, cubic-surface, root-system matroid, classifier, or tiling route. The projective E_7 row is included as a compact control-plus-separator. The 63 projective root pairs have Gram-square and rank-two flat-size pair channels whose full automorphism groups already agree with the 1451520-element projective source action. Nevertheless, embedded row placement is still visible: two A_5 parabolic rows of the same abstract type and order 720 have different centered ranks, $5 = (3,2)$ and $6 = (3,3)$, on $U_{27} \oplus U_{35}$. This is not a standalone E_7 theorem, not a reflection-subgroup classification, not a root-system matroid novelty claim, and not an E_8 or Type-E family theorem. The projective E_8 row is included as the compact endpoint of this E-series comparison. The 120 projective root pairs have Gram-square and rank-two flat-size pair channels whose full automorphism groups already agree with the 348364800-element projective source action. Standard simple-parabolic rows of the same abstract type and order fuse under this action, but source-aware subsystem rows can still separate placement: an A_7 parabolic row and an A_7 row inside a coordinate D_8 subsystem have orbit profiles $[8, 28, 28, 56]$ and $[1, 28, 28, 28, 35]$, with centered ranks $3 = (1, 2)$ and $4 = (1, 3)$ on $U_{35} \oplus U_{84}$. This is not a standalone E_8 theorem, not a Type-E family theorem, and not a root-subsystem classification.

Contributions. The manuscript makes four bounded contributions.

(i) It fixes a five-slot finite-shadow grammar $\mathcal{S} = (N, X, I, \rho, \Phi)$, distinguishes family rank from ambient degree, and gives an admission table for the current FCIG source rows.

(ii) It uses the elementary standard-block dictionary as an interface lens: position-value balance is exactly the vanishing of the $(N-1, 1)$ Specht block, and this lens is applied to both the Type-A row and a signed Type-B/C mirror-union row.

(iii) It gives worked finite examples from the Type-A, modular centered, signed Type-B/C, dihedral, length-colored G_2, projective F_4, projective H_3, projective H_4, projective D_5, projective E_6, projective E_7, projective E_8, and centered-operator artifacts where Φ or the source-channel grammar sees data not determined by coarse histograms, rational rank alone, characteristic-zero data alone, uncolored ambient conjugacy alone, or the finite set alone.

(iv) It records the guardrails: Φ is descriptive, lattice-qualified where Smith form is used, and never used here as a tiling test or as a substitute for the separate theorem routes.

2 The Finite-Shadow Grammar

Definition 2.1 (finite-shadow grammar instance)**.** A finite-shadow grammar instance is a tuple

$$\mathcal{S} = (N, X, I, \rho, \Phi).$$

Here N records the ambient symmetric degree, $X \subseteq S_N$ is the finite row under study, I is the incidence or selection rule, ρ is the representation or centered-operator channel, and Φ is the recorded fingerprint package.

The labels X01–X26 name source families, not the subset X of a single instance. If α is such a family label, we write

$$\mathfrak{F}_\alpha = \{\mathcal{S}_{\alpha,t} : t \in T_\alpha\}, \qquad \mathcal{S}_{\alpha,t} = (N_{\alpha,t}, X_{\alpha,t}, I_{\alpha,t}, \rho_{\alpha,t}, \Phi_{\alpha,t}).$$

A family may have its own rank parameter n; this is distinct from the ambient permutation degree N. Thus a signed rank-n family may act through S_{2n} without identifying its rank with the ambient degree.

The five slots remain the mathematical core. Every admitted concrete instance is governed by the typed *source-lock schema* published with this paper: a provenance and interpretation record fixing the carrier, ambient action and action side, source version, admission, claim owner, verification route, and import/nonimport boundary. Its values are distributed across the canonical registry, the relevant manuscript row, and its cited source or manifest-pinned artifact, as routed by the public crosswalk; they need not be duplicated in one consolidated object. Source partition or layer, coefficient ring, lattice basis, characteristic, quotient/coupling data, placement, and a decision certificate are required exactly when their schema trigger is active. This record prevents the same five formal slots from conflating different source rows; it is not a sixth mathematical slot and does not assert a category of FCIG objects.

For signed rows the ambient object may be a source-locked finite shadow $H \leq S_N$, with $X \subseteq H$. The signed Type-B/C row used later has this form with $B_n \leq S_{2n}$, following the signed-shadow conventions used in the companion Type-B/C artifact. The dihedral row uses the vertex action $I_2(m) \leq S_m$. It is admitted only as a bounded rank-two reflection-shadow exhibit, not as a general Weyl-group row.

Some rows also carry a finite source partition $\Omega = \bigsqcup_i \Omega_i$. This is still a finite-shadow row, but the channel is source-aware: centering may be global on Ω, orbit-centered on the blocks Ω_i, or both, according to the source lock. The length-colored G_2 root-shadow row uses $\Omega = \Omega_{\text{short}} \sqcup \Omega_{\text{long}}$. Forgetting this partition turns the row into an uncolored permutation problem and erases the distinction that the grammar is meant to retain. A source lock can also specify which symmetry layer is admitted on the same finite set. The projective F_4 root-shadow row in Section 9 is used only in this sense: the geometric projective Weyl layer and the source matroid automorphism layer act on the same 24 points, but they give different centered-rank readings because one preserves the length blocks and the other includes a length-swap coset. This is a local source-layer lesson, not a general Weyl theorem.

The projective H_3 row in Section 10 is another same-carrier test, but the active split is not long/short length. The 15 projective root pairs carry a flat-size source channel and a Gram-angle source channel. Their automorphism rows have orders 120 and 60, respectively, while the ordinary standard block is blind. Thus the grammar records not only the finite carrier but also the source channel through which the carrier is read.

The projective E_6 row in Section 13 is a simply laced source-aware control. Unlike the D_5 row, its pair, flat-size, and circuit channels all recover the same projective source action. Its grammar contribution is instead the decomposition of $\mathbb{Q}^{\Pi_{36}}$ into the constant line and two nontrivial source summands U_{20} and U_{15}, which can separate rows with the same total standard rank. The projective E_7 row in Section 14 is a source-exact control with an embedded-parabolic separator. Its Gram-square and rank-two flat-size pair channels already recover the projective source action on Π_{63}, but two embedded A_5 parabolic rows of the same abstract type and order have different (U_{27}, U_{35}) centered rank splits. The projective E_8 row in Section 15 is a source-exact control with a different separator shape. Standard simple-parabolic placements of the same abstract type fuse under the projective Weyl action, but an A_7 parabolic row and an A_7 row inside a coordinate D_8 subsystem have different orbit profiles and (U_{35}, U_{84}) rank splits.

Definition 2.2 (admission, pointer status, and descriptive role)**.** Canonical admission is the two-valued field *admitted* or *not admitted*. Independently, pointer status is *not applicable*, *future*, or *public companion*. Terms such as *core*, *core artifact*, *core exhibit*, *prose*, *meta*, and *companion pointer* are descriptive registry roles, not admission types or new mathematical structures. The separation prevents an unfinished or companion-owned object from entering the manuscript as if it were an admitted theorem source.

The full instance matrix is recorded in Section A. The table is part of the grammar: it says not only what enters, but also what remains outside. This is why the future P18 and P17 companion pointers, the companion-owned exceptional S_6 certificate routes, and the separate public signed-reversal theorem route are not silently absorbed into the present paper.

Remark 2.3 (the teeth test)**.** A proposed FCIG row fails the teeth test if it lacks any of the five slots. A geometric picture without a finite S_N shadow, a finite set without an incidence rule, or a representation calculation without a source-locked finite object can still be useful research material, but it is not a core row of this paper.

3 The Fingerprint Channel Φ

For $X \subseteq S_N$, the fingerprint channel records block data attached to the Specht/Fourier transforms

$$M_\lambda(X) = \sum_{x \in X} D^\lambda(x),$$

where D^λ is a fixed integral Specht model for the partition $\lambda \vdash N$. The background representation theory is standard[Dia88, JK81].

Definition 3.1 (the Φ-package)**.** The package $\Phi(X)$ may include rational ranks $\operatorname{rank}_{\mathbb{Q}} M_\lambda(X)$, the lattice-qualified invariant $\mathrm{SNF}^L_\lambda(X)$, selected finite-field ranks, block-energy data, transport metadata, and rank mass. Every Smith normal form entry is lattice-qualified; no bare Smith form is used as an invariant independent of the chosen integral model. The use of Smith normal form is standard combinatorial linear algebra[Sta16]; the superscript in SNF^L records that the lattice has been fixed.

The following elementary dictionary is the main reason the grammar is useful rather than decorative. It connects the centered-operator language to a specific Specht block.

Proposition 3.1 (standard-block balance dictionary)**.** *Let $X \subseteq S_N$, and let P_x be the permutation matrix of x on $\mathbb{C}^N$, with rows indexed by values and columns by positions. Put*

$$N_X = \sum_{x \in X} P_x, \qquad E_X = N_X - \frac{|X|}{N} J_N,$$

where J_N is the all-ones matrix. Then the following are equivalent:

(a) *X is position-value balanced, i.e. $N_X = (|X|/N)J_N$;*

(b) *E_X vanishes on the standard subspace $V_{\mathrm{std}} = \{v : \sum_i v_i = 0\}$;*

(c) *the $(N-1, 1)$ Specht block $M_{(N-1,1)}(X)$ vanishes, after choosing the corresponding standard model.*

Proof. The permutation representation decomposes as $\mathbb{C}^N = \mathbb{C}\mathbf{1} \oplus V_{\mathrm{std}}$. The constant line records the total row and column sums, while the quotient block is the Specht module of shape $(N-1, 1)$. Subtracting $(|X|/N)J_N$ removes the constant-line part. Thus the centered matrix vanishes exactly when its standard block vanishes. □

Proposition 3.2 (classical left-ideal reading)**.** *Let $d_\lambda = \dim S^\lambda$. For $X \subseteq S_N$, the rank mass used in Φ satisfies*

$$\mathrm{rank\_mass}(X) = \sum_{\lambda \vdash N} d_\lambda \operatorname{rank}_{\mathbb{Q}} M_\lambda(X) = \dim_{\mathbb{C}}(\mathbb{C}[S_N]\mathbf{1}_X),$$

where $\mathbf{1}_X = \sum_{x \in X} x \in \mathbb{C}[S_N]$. This is the standard Wedderburn/Fourier interpretation of the left ideal generated by the indicator element[Ser77, Dia88].

Remark 3.2. Proposition 3.2 is background, not a novelty claim. Its role is to make the fingerprint channel auditable: when rank mass compresses or fails to compress, the statement is a statement about a classical group-algebra left ideal.

4 Modular Centered Channels

The grammar also needs a modular row. Over characteristic zero, a centered constant-line-sum operator is usually read through its rational standard block. Over a field of characteristic p, that is not enough: the same integral operator can have a different finite-field rank after reduction. The source-locked modular row promotes the reserved row X16 from a future placeholder to a compact exhibit.

For a carrier of size N and a field K, let

$$A_N = \{v \in K^N : \sum_i v_i = 0\}.$$

If $p = \operatorname{char} K$ divides N, then the constant line $K\mathbf{1}$ lies inside A_N, so one must also record the reduced quotient $A_N/K\mathbf{1}$. For an integral operator in a fixed anchored augmentation lattice, the Smith data must be lattice-qualified, written SNF^L, before it is compared across characteristics.

Proposition 4.1 (lattice-qualified modular rank bridge)**.** *Fix an integral matrix M in a chosen augmentation-lattice basis, and let*

$$\mathrm{SNF}^L(M) = (d_1, \ldots, d_r, 0, \ldots, 0)$$

be its Smith normal form in that fixed lattice. For every prime p, the rank of the reduced matrix over $\mathbb{F}_p$ is the number of nonzero invariant factors not divisible by p:

$$\operatorname{rank}_{\mathbb{F}_p}(M) = \#\{i : d_i \neq 0 \text{ and } p \nmid d_i\}.$$

Proof. Smith normal form writes M as UDV with U, V unimodular over $\mathbb{Z}$. Reducing modulo p preserves invertibility of U and V over $\mathbb{F}_p$. Hence the finite-field rank is the number of diagonal entries of D that remain nonzero modulo p. The conclusion depends on the chosen integral lattice, which is why the invariant is recorded as SNF^L. $\square$

The point is not the Smith-form fact itself, which is standard [Sta16, Sin13]. The point is that the finite-shadow row has to remember which lattice and which characteristic are being used. The bounded Type-A scout gives a clean witness: the one-incidence layer $I_7(1,1)$ has rational augmentation rank 3, but over $\mathbb{F}_2$ the augmentation rank is 0 even though 2 does not divide 7. This is integral torsion in the operator, not merely the bad-characteristic nesting $K\mathbf{1} \subset A_N$.

Table 1: Modular centered-channel witnesses recorded by the X16 source row.

role	row	p	SNF^L	$\operatorname{rank}_{\mathbb{Q}}$	$\operatorname{rank}_{\mathbb{F}_p}$	reading
carrier-coprime torsion collapse ($2 \nmid 7$)	$I_7(1,1)$	2	$[4,4,8]$	3	0	modular rank sees torsion invisible to rational rank
max bounded scout drop	$I_8(4,1)$	3	$[12^3, 72, 576^3]$	7	0	full augmentation collapse in the bounded replay
quotient drop without augmentation drop	$I_3(0,1)$	3	$[1,1]$	2	2	reduced quotient drops to rank 1 when p divides N
augmentation drop; separate quotient	$I_6(0,4)$	2	$[1,1,8,48,48]$	5	2	augmentation torsion does not determine the quotient channel alone

Thus the modular row contributes a different kind of grammar data from the rational rows above. The row records characteristic, augmentation channel, reduced quotient, and lattice-qualified SNF^L. It does not claim an all-layer Type-A modular theorem, does not import Latin/Sudoku/M19 mechanisms, and is not a modular Specht or tiling theorem.

5 Type-A Source Rows

The Type-A source row is anchored by the published standard-space paper[Bab26f]. Its one-incidence layer I_n is the first mature FCIG row in the grammar. The public Type-A poset-cone and extension-tiler paper supplies the separate chamber/tiler context for rows X01–X02[Bab26l]. The underlying dictionary between poset cones and chamber unions of the braid arrangement is classical; poset cones and their Whitney-number refinements are developed in [DBKR21], and [AM17] treats the faces and cones of real hyperplane arrangements in general. These are context citations only; no theorem from them is imported. This paper uses these definitions and boundaries by citation; it does not merge their theorem claims.

In the notation of Proposition 3.1, the Type-A balance theorem says that for even $n \geq 4$, the one-incidence layer has

$$E_{I_n} = N_{I_n} - \frac{|I_n|}{n} J_n = 0,$$

so the standard block $M_{(n-1,1)}(I_n)$ vanishes. This is not merely vocabulary: it is the exact point where the centered-operator language and the Specht block language meet.

The same row can be replayed through the Fourier/Specht fingerprint artifact. The left-ideal checks below are finite support rows, not new counting theorems. Their exact partition ranks and weighted sums are recorded in the manifest-pinned static certificate `artifacts/x13_type_a_lattice_rank_mass_certificate.json`; the companion read-only checker performs 61 exact checks without creating a thirteenth quantitative replay route.

Table 2: Type-A sample rows from the Fourier/Specht left-ideal closeout.

row	ambient	$\|X\|$	rank_mass(X)
$I_4(1,1)$	S_4	8	6
$I_4(0,2)$	S_4	4	18
$I_5(1,1)$	S_5	20	41
$I_5(2,2)$	S_5	8	30

A second Type-A support row shows why Φ cannot be reduced to a conjugacy-class histogram. The Type-A value witness gives two explicit subsets of S_4, in one-line notation,

$$X_A = \{2314, 2341, 2431, 4231\}, \qquad X_B = \{2413, 2431, 4213, 4231\}.$$

They have the same class histogram $(4) : 1, (3,1) : 2, (2,1,1) : 1$, but the $(3,1)$ block separates them.

For this witness the integral model is explicitly

$$L^{A_3}_{(3,1)} = \mathbb{Z}(e_1 - e_4) \oplus \mathbb{Z}(e_2 - e_4) \oplus \mathbb{Z}(e_3 - e_4),$$

with $D(w)(e_i - e_4) = e_{w(i)} - e_{w(4)}$ and matrices acting on columns. Thus every Smith form in the following table is qualified by this displayed lattice. The static certificate records the two block matrices, determinantal divisors, ranks, and Smith factors.

Table 3: Type-A value witness: same class histogram, different Φ-data.

row	$\mathrm{rank}_{\mathbb{Q}} M_{(3,1)}$	$\mathrm{SNF}^{L}_{(3,1)}$
X_A	2	$(1,2,0)$
X_B	1	$(2,0,0)$

The S_4 witness layer[Bab26i] supplies a smaller finite source row for structured subsets and extension-set comparisons. It is used here as context for how finite rows enter the grammar, not as a license to broaden the Type-A theorem boundary. One notation warning is needed: the public poset-cone paper uses a label P038 for a proved extension-tiler witness, while X10 names the exceptional six-element benchmark family. The bounded P07 syntheme result and P33 six-element certificate are companion-owned routes[Bab26k, Bab26a]; P13 imports neither theorem nor proof, makes no all-n tiling claim, and claims no all-n classification.

6 Signed Type-B/C Source Rows

The signed finite rows come from the companion Type-B/C source package. For the all-n balance statement below, however, we give the elementary proof in full. Write a signed permutation as $g = (\pi, \varepsilon) \in B_n = C_2 \wr S_n$ and embed $B_n \leq S_{2n}$ through its action on $\{+1, \dots, +n, -1, \dots, -n\}$, where $g(+i) = \varepsilon_i(+\pi(i))$ and $g(-i) = \varepsilon_i(-\pi(i))$. Standard Coxeter background for signed permutations is classical[BB05].

The signed incidence row $BI_n(k, l)$ consists of those $g = (\pi, \varepsilon)$ with exactly k indices satisfying $\pi(i) = i, \varepsilon_i = +1$ and exactly l satisfying $\pi(i) = i, \varepsilon_i = -1$. It is a union of B_n-conjugacy classes. The signed artifact verifies that the same fingerprint channel can process six selected rows through the ambient symmetric group. Those six numerical rows and the human-proof all-n proposition below have separate verification postures.

Table 4: Signed finite-shadow rows recorded in the signed replay.

row	ambient	$\lvert X\rvert$	rank_mass(X)
$BI_3(0,0)$	S_6	16	90
$BI_3(0,1)$	S_6	12	300
$BI_3(1,0)$	S_6	12	300
hash-ordered size-$\lvert B_3\rvert$ control	S_6	48	715
$BI_4(0,1)$	S_8	64	10710
$BI_4(1,1)$	S_8	48	17220

The fourth row of Table 4 is not an unspecified random draw: it is the deterministic hash-ordered size-B_3 control recorded by the signed artifact. Its role is only to show that the signed rows are not being compared against a moving baseline.

The first place where Proposition 3.1 genuinely crosses type is the signed mirror-union. Put

$$Y_n^B = BI_n(1,0) \sqcup BI_n(0,1) \subseteq B_n \leq S_{2n}.$$

Let D_m denote the number of derangements of m letters.

Proposition 6.1 (signed mirror-union balance source row)**.** *For $n \geq 3$, the signed mirror-union satisfies*

$$|Y_n^B| = n2^n D_{n-1}, \qquad N_{Y_n^B} = 2^{n-1} D_{n-1} J_{2n}.$$

Equivalently, the ambient standard block vanishes:

$$M_{(2n-1,1)}(Y_n^B) = 0.$$

Proof. Put $f_i^+ = e_{+i} + e_{-i}$ and $f_i^- = e_{+i} - e_{-i}$. The ambient permutation module splits as the B_n-stable direct sum

$$\mathbb{C}^{2n} = U_+ \oplus U_-, \qquad gf_i^+ = f_{\pi(i)}^+, \qquad gf_i^- = \varepsilon_i f_{\pi(i)}^-.$$

Here $U_+ = \mathbb{C}\mathbf{1} \oplus V_{\text{pair}}$, where V_{pair} is the inflated S_n-standard module of dimension $n-1$, while $U_- = V_{\text{ref}}$ is the n-dimensional reflection module. Hence the ambient S_{2n}-standard module restricts to $V_{\text{pair}} \oplus V_{\text{ref}}$.

Let $T_{k,l} = \sum_{g \in BI_n(k,l)} g$. Since the layer is conjugacy invariant, $T_{k,l}$ acts by scalars on these two irreducibles. On every element of the layer the characters of U_+ and U_- are respectively $k+l$ and $k-l$. Taking traces, and subtracting the constant line from U_+, gives

$$\omega_{\text{pair}}(T_{k,l}) = \frac{|BI_n(k,l)|(k+l-1)}{n-1}, \qquad \omega_{\text{ref}}(T_{k,l}) = \frac{|BI_n(k,l)|(k-l)}{n}.$$

For both $(k,l) = (1,0)$ and $(0,1)$, the pair-standard scalar is zero. Moreover

$$|BI_n(1,0)| = |BI_n(0,1)| = n2^{n-1}D_{n-1} :$$

choose the fixed coordinate, derange the other $n-1$ coordinates, and choose their signs. The two reflection scalars are therefore equal and opposite. Their union kills $V_{\text{pair}} \oplus V_{\text{ref}}$, so Proposition 3.1 gives $M_{(2n-1,1)}(Y_n^B) = 0$ and a constant position–value matrix. Finally

$$|Y_n^B| = 2n2^{n-1}D_{n-1} = n2^n D_{n-1}, \qquad \frac{|Y_n^B|}{2n} = 2^{n-1}D_{n-1},$$

which proves the displayed formula for $N_{Y_n^B}$. □

For example, at $n = 3, 4, 5, 6$ the signed-balance check gives $|Y_n^B| = 24, 128, 1440, 16896$ and constants $4, 16, 144, 1408$, respectively. This is the concrete signed use of the standard-block dictionary: Type A balances a single one-incidence layer in its mature theorem row, while the signed row balances a mirror-union because the ambient standard splits into two signed channels.

The row $BI_3(0,0)$ compresses to the sign-subgroup index background. The mixed rows, such as $BI_3(0,1)$ and $BI_4(0,1)$, are the bounded signal for this paper: they are neither random-sized full support nor a promoted closed formula. They show that the finite-shadow interface sees signed incidence structure while keeping the signed artifact as a bounded dependency.

The signed value witness gives the signed analogue of the Type-A value witness. In one-line notation on S_6,

$$X_A = \{123465, 124365\}, \qquad X_B = \{123465, 214356\}.$$

Equivalently, inside B_3, these are $X_A = s_3\langle s_2\rangle$ and $X_B = s_3\langle s_1s_2s_3\rangle$. They have the same signed-cycle histogram and the same rank mass, but the standard block over S_6 separates them.

Table 5: Signed value witness: same signed histogram, different standard block.

row	rank_mass	$\text{rank}_{\mathbb{Q}} M_{(5,1)}$	$\text{SNF}^L_{(5,1)}$
X_A	360	4	$(1,2,2,2,0)$
X_B	360	2	$(1,1,0,0,0)$

No broader Coxeter-group or classification statement is made in this section. The six rank-mass rows remain bounded P14 artifact data; the mirror-union proposition is the separate P13-owned elementary human-proof claim X15-002. Neither imports the P15 signed-reversal theorem.

7 Dihedral Vertex-Shadow Source Rows

The third worked finite-shadow family is the vertex action of the dihedral group

$$W_m = I_2(m) = \langle r, s \mid r^m = s^2 = 1,\ srs = r^{-1}\rangle \le S_m,$$

where the vertices are indexed by $\mathbb{Z}/m\mathbb{Z}$, $r(i) = i+1$, and $s_a(i) = a - i$. This is standard dihedral/Coxeter material[BB05]. The FCIG role is narrower: the row tests the same centered standard channel and the same lattice-qualified fingerprint discipline on a rank-two reflection shadow.

The source-locked rows are

$$C_m = \{r^a : a \in \mathbb{Z}/m\mathbb{Z}\}, \qquad F_m = \{s_a : a \in \mathbb{Z}/m\mathbb{Z}\}, \qquad X_{m,a} = \{s_0, s_a\}.$$

For even m, the half-reflection rows F_m^0 and F_m^1 consist of the reflections with even and odd parameters.

Proposition 7.1 (dihedral standard-block rows)**.** *Over characteristic zero, the centered standard ranks of the dihedral vertex-shadow rows satisfy the following.*

(a) *C_m and F_m are position-value balanced, so their centered standard rank is* 0.

(b) *If m is even, F_m^0 and F_m^1 have centered standard rank* 1.

(c) *For $1 \le a < m$, put $d = \gcd(a, m)$ and $L = m/d$. Then*

$$\operatorname{rank}_{\mathrm{std}}(X_{m,a}) = (m-1) - d\,\mathbf{1}[L \textit{ is even}].$$

Source-locked proof sketch. The rotation row and the all-reflection row are sharply transitive on vertices, so their position-value matrices are J_m. For a half-reflection row, the parity vector $((-1)^i)_i$ gives the only centered defect, hence rank 1. Finally, after multiplying by the invertible reflection s_0, the two-mirror standard operator is equivalent to $I + r^a$ on V_{std}. The rotation r^a has d cycles of length $L = m/d$, and $I + r^a$ has an alternating kernel exactly on the even cycles. □

A more uniform way to see Proposition 7.1 is the reflection-mask Fourier description. For $A \subseteq \mathbb{Z}/m\mathbb{Z}$, define $R_A = \{r^a : a \in A\}$, $F_A = \{s_a : a \in A\}$, and $p_A(t) = \sum_{a\in A} t^a$. On the centered standard block,

$$\operatorname{rank}_{\mathrm{std}}(R_A) = \operatorname{rank}_{\mathrm{std}}(F_A) = \#\{1 \le j \le m-1 : p_A(\zeta_m^j) \ne 0\}.$$

Since

$$1 + t + \cdots + t^{m-1} = \prod_{\substack{d|m\\ d>1}} \Phi_d(t),$$

the rational rank defect is

$$\sum_{\substack{d|m,\ d>1\\ \Phi_d(t)|p_A(t)}} \varphi(d),$$

so the finite dihedral rank defects are exactly cyclotomic resonances of the mask polynomial. Equivalently, over $\mathbb{Q}$, this is $(m-1) - \deg\gcd(p_A(t), 1 + t + \cdots + t^{m-1})$. This is classical cyclic Fourier algebra used here as an explanation, not as a new Fourier theorem.

The lattice-qualified two-mirror statement is the most useful dihedral payoff for the grammar. On the standard lattice A_{m-1}, with $d = \gcd(a, m)$ and $L = m/d$,

$$\mathrm{SNF}^L_{\mathrm{std}}(X_{m,a}) = \begin{cases} 1^{m-1-d}, 0^d, & L \text{ even},\\ 1^{m-d}, 2^{d-1}, & L \text{ odd}. \end{cases}$$

Thus rational rank is only the shadow of a stronger lattice fingerprint.

The common-precedence closure gives a short bridge back to the Type-A chamber language, but it is controlled by a different parent datum. For a nonempty mask $A = \{a_0, \ldots, a_{k-1}\} \subseteq \mathbb{Z}/m\mathbb{Z}$, let $g_1, \ldots, g_k$ be the cyclic gap lengths between consecutive selected cutpoints. The

Table 6: Dihedral vertex-shadow witnesses.

witness	coarse data held fixed	fingerprint separation
$X_{8,2}$ vs. $X_{8,4}$	both rows contain two even reflections, hence the same elementwise dihedral reflection-class histogram	centered standard ranks 5 and 3
$X_{9,1}$ vs. $X_{9,3}$	both rows have two reflections and full rational standard rank 8	$\mathrm{SNF}^L_{\mathrm{std}}(X_{9,1}) = 1^8$, while $\mathrm{SNF}^L_{\mathrm{std}}(X_{9,3}) = 1^6, 2^2$

common-precedence poset of R_A is the disjoint union of cyclic gap chains of lengths $g_1, \dots, g_k$, while F_A has the same description after label negation. Hence the closure size is

$$\frac{m!}{g_1! \cdots g_k!}.$$

This cut/gap channel is not determined by rational rank or SNF^L alone. The named rows specialize as follows: C_m and F_m have empty common-precedence relation and close to all of S_m; for $m = 2q$, each half-reflection row closes to q disjoint two-element chains, with $(2q)!/2^q$ linear extensions; and $X_{m,a}$ closes to the shuffle of two chains of lengths a and $m - a$, with $\binom{m}{a}$ linear extensions. This is only a closure diagnostic; it is not an exact-poset-cone theorem, not a tiling theorem, and not a classifier.

The boundary is the same as in the source artifact. This section makes no standalone dihedral paper claim, no new Fourier theory claim, no G_2 claim from the special case $I_2(6)$, no general Weyl-group theorem, and no rank-implies-tiling statement.

8 A Length-Colored G_2 Root Shadow

The dihedral row of Proposition 7.1 is an uncolored vertex-shadow row. The bounded G_2 row records a different source feature. The finite set is the twelve-root shadow

$$\Omega = \Omega_{\mathrm{s}} \sqcup \Omega_{\mathrm{l}}, \qquad |\Omega_{\mathrm{s}}| = |\Omega_{\mathrm{l}}| = 6,$$

where the two six-cycles are the short and long roots. The Weyl group $W(G_2)$ acts on Ω and preserves the two length blocks. This section uses only that finite action and the elementary rank-two Coxeter background[BB05]. It is not a general Weyl-group theorem.

Let $\mathbf{1}_{\mathrm{s}}$ and $\mathbf{1}_{\mathrm{l}}$ be the two block indicators, and put

$$u_{\mathrm{len}} = \mathbf{1}_{\mathrm{s}} - \mathbf{1}_{\mathrm{l}}.$$

The centered permutation module decomposes as

$$\mathbb{Q}^{\Omega} = \mathbb{Q}\mathbf{1} \oplus \mathbb{Q}u_{\mathrm{len}} \oplus V_{\mathrm{s}}^0 \oplus V_{\mathrm{l}}^0.$$

The line $\mathbb{Q}u_{\mathrm{len}}$ is the obstruction that does not exist in the uncolored dihedral vertex row.

Proposition 8.1 (source-channel ranks in the length-colored G_2 shadow)**.** *For the source-locked G_2 root shadow above, the following bounded statements hold.*

(a) *Global centered balance on all twelve roots is blocked by the length line $\mathbb{Q}u_{\mathrm{len}}$, because every admitted row preserves short and long roots separately.*

(b) *After orbit-centering on Ω_s and Ω_l, the full Weyl row, the rotation row, and the full reflection row are balanced on both blocks. The short-axis and long-axis half-reflection rows have orbit-centered rank* 2*, one rank-one defect on each length block.*

(c) *For two-mirror rows $X_a = \{\tau_0, \tau_a\}$, the restrictions to the short and long blocks are two copies of the dihedral $m = 6$ row. Hence*

$$\mathrm{rank}_{\mathrm{orbit}}(X_a) = 2r(a), \qquad \mathrm{rank}_{\mathrm{global}}(X_a) = 1 + 2r(a),$$

where $r(a)$ is the dihedral centered rank from Proposition 7.1 for $m = 6$.

Source-locked proof sketch. The first assertion is the source-partition obstruction: each permutation matrix is block diagonal with respect to $\Omega_\mathrm{s} \sqcup \Omega_\mathrm{l}$, so the length line survives global centering. The orbit-centered statements remove the two block constants separately; on each six-cycle the action reduces to the dihedral calculation already recorded in Proposition 7.1. A half-reflection row fixes one parity pattern on the short block and one on the long block, giving two independent rank-one defects. A two-mirror row restricts to two identical copies of the $m = 6$ two-mirror calculation, and the global channel adds back exactly the length line. □

The source partition is also necessary for comparison. If the length labels are forgotten, a short-axis half-reflection row and a long-axis half-reflection row are ambient-conjugate inside the uncolored symmetric group on Ω. The source profile distinguishes them: one fixes short roots on its axis and the other fixes long roots. Thus uncolored ambient conjugacy is too coarse for this row, while the source-aware grammar keeps the data that the finite shadow came with.

Table 7: What the length-colored G_2 row contributes to the grammar.

feature	finite fact	grammar reading
length line	u_{len} survives global centering	source partitions can be genuine channels, not annotations
orbit-centered balance	full Weyl, rotation, and reflection rows balance on each length block	centering must sometimes be blockwise rather than global
two-mirror transfer	ranks are two copies of the $m = 6$ dihedral row plus the length line globally	the dihedral transfer only applies after the source blocks are kept
source-profile witness	short-axis and long-axis rows are ambient-conjugate after forgetting length	uncolored symmetric-group conjugacy can erase source information

The boundary is strict. This is a length-colored finite-shadow exhibit, not a theorem about arbitrary Weyl groups, affine arrangements, or projective quotients. The uncolored dodecagon is a useful control, but it is not a replacement source for the G_2 row, because it removes exactly the short/long channel that explains the ranks above.

9 A Source-Aware Projective F_4 Root Shadow

The length-colored G_2 row shows that source partitions can be active channels. The projective F_4 row records a slightly different source-aware phenomenon: on the same finite set, changing the admitted source layer changes the centered-rank reading.

Let Π_{24} be the 24 antipodal pairs of F_4 roots, split into long and short projective roots

$$\Pi_{24} = L \sqcup S, \qquad |L| = |S| = 12,$$

and put $u_{\mathrm{len}} = \mathbf{1}_L - \mathbf{1}_S$. The projective image G_{proj} of $W(F_4)$ has order 576 and preserves the two length blocks. The source matroid layer is the automorphism group of the 24-point F_4 root-pair matroid studied by Fried–Gerek–Gordon–Perunicic[FGGP07]. In the source-locked notation used here it is

$$G_{\mathrm{mat}} = \langle G_{\mathrm{proj}}, \sigma\rangle, \qquad |G_{\mathrm{mat}}| = 1152,$$

where σ is a non-geometric matroid automorphism that swaps the two length blocks. The automorphism order is used here as a source fact and is also verified in the companion source package from the rank-two flat hypergraph of the represented 24-point matroid; this paper does not claim to discover the F_4 matroid automorphism group.

Proposition 9.1 (source-layer ranks in the projective F_4 shadow)**.** *For the source-locked projective F_4 root shadow above, the following bounded statements hold.*

(a) *The geometric row G_{proj} has global centered rank* 1*. More precisely, if $N_X = \sum_{g\in X} P_g$ and $E_X = N_X - (|X|/24)J$, then*

$$E_{G_{\mathrm{proj}}} = 24\, u_{\mathrm{len}} u_{\mathrm{len}}^{\mathsf{T}}.$$

After centering separately on L and S, this residual is zero.

(b) *The matroid source row G_{mat} is globally balanced on the same 24 points: $E_{G_{\mathrm{mat}}} = 0$. Equivalently, the length-swap coset cancels the long-minus-short residual.*

(c) *The six-block source quotient gives no independent quotient residual after six-block centering. The two nonidentity block-kernel conjugacy rows K_{24} and K_{16} are finite supporting witnesses, each with six-block-centered rank* 18*. The subscripts record moved-point support sizes, not row cardinalities; the certified rows have* 6 *and* 9 *elements, respectively.*

(d) *The rank-two flat closure-size relation is not determined by the long/short partition; its closure-value matrix has length-centered residual rank* 22 *and six-block residual rank* 18*. This is recorded as an incidence invariant of the source, not as an additional row classifier.*

Source-locked proof sketch. The first two assertions are the two-block centering calculation. A group acting transitively on each of two equal blocks and preserving the blocks has orbit sum $(|G_{\mathrm{proj}}|/12)(J_L + J_S) = 48(J_L + J_S)$; subtracting the global center $24J$ gives $24u_{\mathrm{len}}u_{\mathrm{len}}^{\mathsf{T}}$. Adding the source length-swap coset gives $48J$, which is exactly the global center for the row of size 1152. The matroid-layer statement is source-locked by [FGGP07] and replayed by reconstructing the automorphism group from the rank-two flat hypergraph. The block-kernel and flat-incidence statements are finite replay certificates for this 24-point source; they are included only with the boundaries stated in (c) and (d). □

Table 8: Evidence carried by the projective F_4 source-layer exhibit.

layer	row or relation	certified finite fact	grammar reading
geometric row	G_{proj}	global rank 1; length-centered rank 0	one long/short residual line
matroid row	G_{mat}	global rank 0	source length-swap cancellation
block kernel	K_{24}, K_{16}	six-block-centered rank 18	finite kernel-conjugacy witness only
flat relation	rank-two closure size	length residual rank 22	incidence invariant; no extra row witness

Thus the exhibit-scoped lesson is narrow but useful: for this 24-point source, the finite set alone is not the row. This exhibit shows that, for this source, the source partition and the admitted source layer must be recorded before centered-rank data is interpreted. Nothing here is promoted to a full F_4, 24-cell, block/triality, flat-row, Weyl-group, classifier, or tiling theorem.

10 A Projective H_3 Source-Channel Exhibit

The projective F_4 row shows that the admitted source layer can change the centered-rank reading on one finite set. The next exhibit gives the same lesson in a non-crystallographic rank-three source, but with a different channel split: matroid-flat incidence versus Gram-angle geometry.

Let Π_{15} be the 15 antipodal pairs in the H_3 root system, constructed over $\mathbb{Q}(\phi)$ with $\phi^2 = \phi + 1$. The source package uses the simple-root Gram matrix

$$\begin{pmatrix} 2 & -\phi & 0 \\ -\phi & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix},$$

generates the 30 roots, and passes to projective pairs. The rank-two flats have sizes $2, 3, 5$, and the finite proof spine colors the $\binom{15}{2} = 105$ unordered pairs in two ways: by flat size and by squared Gram value. Classical root-system and matroid background is standard; see [BB05, SFT11]. The statements below are used only as a finite source-channel exhibit for this 15-point shadow.

Proposition 10.1 (source channels in the projective H_3 shadow)**.** *For the source-locked projective H_3 shadow Π_{15}, the bundled public replay certifies the following bounded facts.*

(a) *The automorphism group of the flat-size edge coloring has order* 120*; denote this row by* G_{mat}.

(b) *The automorphism group of the Gram edge coloring has order* 60*, equal to the geometric projective row* $G_{\mathrm{geo}} = G_{\mathrm{gram}}$.

(c) *The complementary row* $G_{\mathrm{ng}} = G_{\mathrm{mat}} \setminus G_{\mathrm{geo}}$ *has* 60 *elements. Every element of this row preserves the flat-size channel, and no element preserves the Gram-color channel.*

(d) *The ordinary standard Specht block* $(14, 1)$ *is blind on all four rows* $G_{\mathrm{mat}}, G_{\mathrm{geo}}, G_{\mathrm{ng}}, G_{\mathrm{gram}}$. *Higher ordinary* S_{15} *Specht blocks do see the subgroup difference; the first separator is* $(13, 2)$*, with ranks* $2, 3, 3$ *on* $G_{\mathrm{mat}}, G_{\mathrm{geo}}, G_{\mathrm{ng}}$*, respectively.*

Source-locked proof sketch. The bundled public replay audits an exported finite payload computed exactly over $\mathbb{Q}(\phi)$. It uses the two 105-edge color payloads and recomputes the full color-preserving permutation groups in S_{15}. The flat-size payload has automorphism group G_{mat} of order 120; the Gram payload has automorphism group $G_{\mathrm{geo}} = G_{\mathrm{gram}}$ of order 60. The same replay checks that $G_{\mathrm{mat}} = G_{\mathrm{geo}} \sqcup G_{\mathrm{ng}}$ and that the non-geometric coset contains no Gram-color-preserving element. The same public replay also verifies the displayed ordinary standard and $(13, 2)$ rank readings; the full source package contains the broader ordinary S_{15} rank scout. It is descriptive fingerprint data; the source-channel separation is the preceding color-automorphism calculation. □

A display witness is the involution

$$\tau = \texttt{ACBLMONHKJIDEGF} = (B\ C)(D\ L)(E\ M)(F\ O)(G\ N)(I\ K).$$

It preserves all flat families and flat-size colors, but changes the Gram color of the edge AD from $1 + \phi$ to the color of AL, namely $2 - \phi$. In the source replay, this witness changes Gram colors on 60 edges.

Thus the H_3 row contributes one compact lesson to the grammar: a finite root shadow may be balanced in the ordinary standard layer while its source channels still define different FCIG rows. The exhibit is not a theorem about all H_3 rows, all non-crystallographic Coxeter systems, root-system matroids, classifiers, or tilers.

Table 9: Evidence carried by the projective H_3 source-channel exhibit.

channel	row or diagnostic	certified finite fact	grammar reading
flat-size channel	G_{mat}	automorphism order 120	matroid-flat source symmetry
Gram channel	$G_{\mathrm{geo}} = G_{\mathrm{gram}}$	automorphism order 60	geometric angle symmetry
non-geometric coset	G_{ng}	preserves flats, changes Gram colors	same carrier, different source channel
ordinary standard block	Specht $(14, 1)$	rank 0 on all target rows	negative control: standard layer is blind
higher ordinary block	Specht $(13, 2)$	ranks $2, 3, 3$	descriptive higher fingerprint sees the split

11 A Projective H_4 Source-Channel Exhibit

The projective H_3 row shows that a source channel can separate finite rows even when the ordinary standard block is blind. The final reflection-shadow exhibit uses the 60 antipodal pairs in the H_4 root system, viewed as a projective 600-cell root shadow. Its role is narrower than a theorem about H_4: it compares two source channels on the same carrier of 60 projective points.

Let E_{60} denote those 60 antipodal pairs. The source package uses the rank-two flat-size channel of the root-system matroid and the squared-Gram channel coming from the geometric realization. The background fact that $M(H_4)$ has a matroid automorphism group of order 14400, half geometric and half non-geometric, is classical in this source context; see Bao–Friedman-Gerlicz–Gordon–McGrath–Vega[BFGG+11] and the general root-system matroid background of [SFT11]. The statement below is only a finite replay of the two colored channels inside the present FCIG grammar. The source-context names G_{geo} and G_{mat} come from this source package and the cited matroid literature; the public replay itself certifies the two color-channel automorphism orders and the displayed witness.

Proposition 11.1 (source channels in the projective H_4 shadow)**.** *For the source-locked projective H_4 shadow E_{60}, the bundled H4 replay certifies the following bounded facts.*

(a) *The full automorphism group of the unordered-pair channel colored by rank-two flat size has order* 14400*.*

(b) *The full automorphism group of the unordered-pair channel colored by squared Gram value has order* 7200*.*

(c) *The two channels give different orbit readings on unordered pairs. The flat-size channel has an automorphism carrying* (`E60_00`, `E60_21`) *to* (`E60_00`, `E60_33`)*, while the Gram-square channel has no color-preserving automorphism carrying one pair to the other.*

(d) *The recorded Gram colors of these two pairs are* $1/2| - 1/4t$ *and* $1/4|1/4t$*, respectively. In the source-context reading, this is the finite witness behind the distinction between the matroid/flat-size row and the geometric/Gram row.*

Source-locked proof sketch. The bundled H4 source-channel replay stores the exact finite payload for the two complete colored graphs on the $\binom{60}{2} = 1770$ unordered pairs. It recomputes the full Gram-square color automorphism group and obtains 7200, recomputes the full flat-size color automorphism group by a resolving-base enumeration and obtains 14400, and checks the pair-orbit witness stated above. The replay does not ask the reader to trust an unbundled generator list: it certifies the two colored-channel groups directly from the finite payload. The names G_{geo} and G_{mat} are source-context labels supplied by the cited H4 matroid literature and the source package. These are finite channel computations, not new source theorems about H_4. □

Table 10: Evidence carried by the projective H_4 source-channel exhibit.

channel	certified finite fact	grammar reading
carrier	60 antipodal pairs, 1770 unordered pairs	projective 600-cell root shadow, not the full 120-root set
flat-size channel	full color automorphism group 14400	matroid/source-incidence symmetry
Gram-square channel	full color automorphism group 7200	geometric metric symmetry
pair witness	same flat-size orbit, distinct Gram-square orbits	same carrier, different source channel
pair-orbit counts	4 Gram-square channel orbits versus 3 flat-size channel orbits	source-aware grammar sees the channel split

Thus the H_4 row contributes one compact lesson: a projective root shadow on 60 antipodal root pairs can carry two natural source channels with different full color automorphism groups and different pair-orbit readings. It is not a standalone H_4 paper, not a 600-cell classification, not a Weyl/Coxeter theorem, and not a Φ classifier.

12 A Projective D_5 Root-Shadow Control Exhibit

The projective H_4 row shows that two pair-color channels on the same carrier may have different automorphism groups. The D_5 control below has a different role. It is crystallographic and simply laced, so there is no short/long length split. The useful boundary is instead between pair-level data and rank-two triple incidence: the pair channel is too coarse, while the A_2 3-flat incidence recovers the source projective Weyl action exactly.

Let Π_{20} be the 20 antipodal pairs in the D_5 root system, represented by the projective signed coordinate-edge roots $\{L^+_{ij}, L^-_{ij} : 1 \le i < j \le 5\}$. The source package computes the projective Weyl row $W(D_5)_{\mathrm{proj}}$, the pair-level Gram/flat-size channel on $\binom{20}{2}$ unordered pairs, and the family of projective A_2 rank-two flats. Classical background on finite Coxeter and root systems is standard; see [BB05]. The proposition below is only a finite source-channel control exhibit for this 20-point shadow.

Proposition 12.1 (source recovery from A_2 incidence in the projective D_5 shadow)**.** *For the source-locked projective D_5 shadow Π_{20}, the exact source replay certifies the following bounded facts.*

(a) *The pair-level Gram/flat-size channel has full automorphism group*

$$\mathrm{Aut}(\mathrm{pair}) \cong C_2^{10} : S_5, \qquad |\,\mathrm{Aut}(\mathrm{pair})| = 122880.$$

(b) *The source projective Weyl action has order*

$$|W(D_5)_{\mathrm{proj}}| = 1920.$$

Thus the pair channel has 64 *times too many automorphisms.*

(c) *The 40 projective A_2 3-flat incidences have preservers exactly equal to $W(D_5)_{\mathrm{proj}}$ inside the pair-channel automorphism group.*

(d) *The five coordinate D_4 size-12 flats are a negative control: their preservers inside the pair channel still have order* 122880*, so this localization is sign-blind by itself.*

(e) *The fake flip $L_{12}^+ \leftrightarrow L_{12}^-$ preserves the pair channel and the coordinate D_4 family, is not in $W(D_5)_{\mathrm{proj}}$, and breaks the A_2 flat $\{L_{12}^+, L_{13}^+, L_{23}^-\}$.*

Source-locked proof sketch. The exact replay builds the 20 projective signed-edge labels, computes the projective $W(D_5)$ action, and recomputes the full color-preserving automorphism group of the pair-level channel. The ten twin classes $\{L_{ij}^+, L_{ij}^-\}$ give an independent C_2^{10} flip kernel, while the quotient is the line graph $L(K_5)$ and has automorphism group S_5. This gives $C_2^{10} : S_5$ and order 122880. The same replay enumerates the 40 projective A_2 3-flats and checks that their preservers inside the pair-channel group have order 1920 and equal the computed projective Weyl row. The displayed fake flip is then checked directly against pair data, the D_4 family, the A_2 family, and membership in $W(D_5)_{\mathrm{proj}}$. □

Table 11: Evidence carried by the projective D_5 source-control exhibit.

channel or witness	certified finite fact	grammar reading
carrier	20 projective signed-edge root pairs	finite D_5 root shadow, not a Type-E theorem
pair-level channel	$\lvert \mathrm{Aut(pair)}\rvert = 122880 = C_2^{10} : S_5$	pair data over-automorphizes the source by factor 64
source row	$\lvert W(D_5)_{\mathrm{proj}}\rvert = 1920$	target source action
A_2 3-flat incidence	40 flats; preservers exactly $W(D_5)_{\mathrm{proj}}$	rank-two triple incidence restores the source
coordinate D_4 flats	5 flats; preservers still order 122880	negative control: localization alone is sign-blind
fake flip	$L_{12}^+ \leftrightarrow L_{12}^-$ preserves pair/D4, breaks A_2	explicit boundary witness

Thus the D_5 row contributes one compact control lesson: in a simply laced projective root shadow, pair-level Gram/flat-size data can still be too coarse, and a higher incidence channel may be required to recover the source action. It is not a standalone D_5 paper, not a general Type-E theorem, not an $E_6/E_7/E_8$ transfer, not a matroid classification, not a tiling result, and not a Φ classifier.

13 A Projective E_6 Source-Aware Control Exhibit

The preceding D_5 row shows that pair-level data can be too coarse even in a simply laced projective root shadow. The E_6 row is a complementary control. Here the classical pair, flat, and circuit channels all collapse to the same projective source action, while a source-aware decomposition still detects row information that total standard rank does not.

Let Π_{36} be the 36 antipodal pairs in the E_6 root system. The source package computes the projective Weyl action, the projective root graph Γ_{E_6} whose edges join projective pairs represented by roots α, β with $\lvert\langle \alpha, \beta\rangle\rvert = 1$, the flat-size pair channel, the rank-two circuit channel, and a compact table of parabolic source rows. Classical background on finite Coxeter groups and root-system matroids is standard; see [BB05, SFT11]. The proposition below is only a bounded source-aware control exhibit for this 36-point shadow.

Proposition 13.1 (channel collapse and source-aware split in the projective E_6 shadow)**.** *For the source-locked projective E_6 shadow Π_{36}, the exact source replay certifies the following bounded facts.*

(a) *The carrier has* 72 *roots and* 36 *projective root pairs. The projective Weyl image has order*

$$|W(E_6)_{\mathrm{proj}}| = 51840.$$

(b) *The projective root graph Γ_{E_6} is strongly regular with parameters*

$$(36, 20, 10, 12),$$

and has full automorphism group of order 51840*.*

(c) *The flat-size pair channel and the rank-two circuit channel do not enlarge this row: the flat-size channel collapses to the same projective graph data, the circuit channel has* 120 *triples, and its full automorphism group also has order* 51840*.*

(d) *As a source-aware permutation module, the rational channel is recorded as*

$$\mathbb{Q}^{\Pi_{36}} = \mathbb{Q}\mathbf{1} \oplus U_{20} \oplus U_{15}.$$

The parabolic rows Par_{012} *and* Par_{0135} *both have size* 24 *and total centered standard rank* 8*, but their* (U_{20}, U_{15}) *splits are respectively* $(6, 2)$ *and* $(5, 3)$*.*

(e) *The two displayed rows are not claimed to be same-type or same-orbit indistinguishability witnesses: in the source table they are of types A_3 and $A_2 + A_1 + A_1$, and the source replay records no same-type or same-size/same-orbit split witness.*

Source-locked proof sketch. The exact replay constructs the E_6 root system, quotients by antipodal pairs, computes the induced projective Weyl action, and verifies its order. It then builds the pair graph from the projective root inner-product channel and checks the strongly regular parameters and full automorphism order. The flat-size channel is compared against the same pair data, and the circuit channel is computed as the family of rank-two dependent triples; its automorphism group has the same order. Finally, the parabolic row table records total standard rank together with the two nontrivial source summands U_{20} and U_{15}, producing the displayed A_3 versus $A_2 + A_1 + A_1$ witness and the negative same-type/same-orbit counts. □

Thus the E_6 row contributes one compact control lesson: even when the pair, flat, and circuit channels all recover the same classical projective source group, the source-aware decomposition can still record information hidden by total standard rank. It is not a standalone E_6 paper, not a Type-E theorem, not a Schlaefli/double-six/27-line or cubic-surface route, not a root-system matroid novelty claim, not a classifier, and not a tiling result.

14 A Projective E_7 Source-Exact Control and Embedded-Parabolic Separator

The preceding E_6 row shows that source-aware summands can matter even after the low-level channels collapse to the classical source action. The E_7 row is a sharper control. On the 63 projective root pairs, the pair-level channels already recover the projective source action exactly, but embedded parabolic rows can still separate source placement.

Let Π_{63} be the 63 antipodal pairs in the E_7 root system. The source package constructs the projective Weyl action, the projective root graph Γ_{E_7} from the squared Gram channel, the rank-two flat-size pair channel, the source decomposition of $\mathbb{Q}^{\Pi_{63}}$, and selected simple-parabolic rows. Classical root-system and Coxeter background is standard; see [BB05]. Reflection-subgroup classification and root-system matroid automorphism theory are prior-art contexts, not claims of this exhibit; see [DL11, SFT11].

Table 12: Evidence carried by the projective E_6 source-aware control exhibit.

channel or witness	certified finite fact	grammar reading
carrier	72 roots, 36 projective root pairs	finite simply laced exceptional source shadow
projective source row	$\lvert W(E_6)_{\mathrm{proj}}\rvert = 51840$	target source action
pair/projective graph	Γ_{E_6} has parameters $(36, 20, 10, 12)$, automorphism order 51840	pair channel already matches the source action
flat-size and circuit channels	flat-size collapses to the graph; 120 circuit triples with automorphism order 51840	no extra automorphism enlargement from these channels
source split	$\mathbb{Q}^{\Pi_{36}} = \mathbb{Q}\mathbf{1} \oplus U_{20} \oplus U_{15}$	the grammar keeps source summands, not just total rank
parabolic witness	Par_{012} and Par_{0135}: size 24, rank 8, splits $(6, 2)$ and $(5, 3)$	total standard rank is blind to source-aware row information
negative witness boundary	types A_3 and $A_2 + A_1 + A_1$; same-type and same-size/same-orbit counts are 0	no stronger indistinguishability claim

Proposition 14.1 (source-exact pair channels and embedded parabolic separation in the projective E_7 shadow)**.** *For the source-locked projective E_7 shadow Π_{63}, the exact source replay certifies the following bounded facts.*

(a) *The carrier has* 126 *roots and* 63 *projective root pairs. The projective Weyl image has order*

$$|W(E_7)_{\mathrm{proj}}| = 1451520.$$

(b) *The projective root graph Γ_{E_7} has strongly regular parameters*

$$(63, 32, 16, 16),$$

and its full automorphism group has order 1451520*. The rank-two flat-size pair channel has the same full automorphism group, so the pair/flat level is already source-exact.*

(c) *As a source-aware permutation module,*

$$\mathbb{Q}^{\Pi_{63}} = \mathbb{Q}\mathbf{1} \oplus U_{27} \oplus U_{35}.$$

The reflection row satisfies

$$N_{\mathrm{Ref}} = 31I + A,$$

where A is the adjacency matrix of Γ_{E_7}; on the centered part this gives rank $62 = 27 + 35$, with scalar readings 35 *and* 27 *on the two nontrivial summands.*

(d) *The parabolic rows* Par_{12345} *and* Par_{12347} *have the same abstract type A_5 and the same order* 720*, but different embedded source fingerprints:*

$$\mathrm{rank}_{\mathrm{std}}(\mathrm{Par}_{12345}) = 5 = (3, 2), \qquad \mathrm{rank}_{\mathrm{std}}(\mathrm{Par}_{12347}) = 6 = (3, 3),$$

where the ordered pairs record the (U_{27}, U_{35}) ranks. Their orbit profiles on Π_{63} are respectively

$$[1, 6, 6, 15, 15, 20] \quad \textit{and} \quad [1, 1, 1, 15, 15, 15, 15].$$

(e) *The replay also records smaller support witnesses of the same embedded-placement type, for instance* $A_1 + A_1 + A_1$ *rows* Par_{135} *and* Par_{137}*, and* $A_3 + A_1$ *rows* Par_{1235} *and* Par_{1237}*. These support the bounded reading only; they are not promoted to a parabolic classification.*

(f) *The raw subsystem-reflection rows for the named examples do not provide the separator: they have full centered rank* $62 = 27 + 35$*. Thus the finite witness is the embedded parabolic subgroup row, not merely the set of reflections in the embedded subsystem.*

Source-locked proof sketch. The exact replay constructs the E_7 roots, quotients by antipodal pairs, computes the induced projective source action, and verifies its order. It then constructs Γ_{E_7} from the squared Gram channel, checks the strongly regular parameters, and computes that both the graph channel and the rank-two flat-size pair channel have full automorphism group of order 1451520. The same source package decomposes $\mathbb{Q}^{\Pi_{63}}$ into $\mathbb{Q}\mathbf{1} \oplus U_{27} \oplus U_{35}$, computes the reflection-row identity $N_{\mathrm{Ref}} = 31I + A$, and evaluates the selected simple-parabolic row matrices on the two source summands. The displayed A_5 pair has the same abstract type and order, but different rank and orbit-profile data on the same carrier. The reflection-set negative controls are recomputed separately and have full centered rank. $\square$

Table 13: Evidence carried by the projective E_7 source-exact control and embedded-parabolic separator.

channel or witness	certified finite fact	grammar reading
carrier	126 roots, 63 projective root pairs	simply laced exceptional projective source shadow
projective source row	$\lvert W(E_7)_{\mathrm{proj}}\rvert = 1451520$	target source action
pair/flat channels	Γ_{E_7} has parameters $(63, 32, 16, 16)$; Gram-square and rank-two flat-size pair channels both have automorphism order 1451520	pair-level data is already source-exact
source split	$\mathbb{Q}^{\Pi_{63}} = \mathbb{Q}\mathbf{1} \oplus U_{27} \oplus U_{35}$	row data is read source-summand-wise
reflection row	$N_{\mathrm{Ref}} = 31I + A$, rank $62 = 27 + 35$	source-exact carrier still has meaningful row identities
main parabolic witness	Par_{12345} and Par_{12347}: both A_5, both order 720, ranks $5 = (3, 2)$ and $6 = (3, 3)$	same abstract parabolic type/order does not determine embedded FCIG fingerprint
support witnesses	A_1^3: ranks $25 = (12, 13)$, $26 = (12, 14)$; $A_3 + A_1$: ranks $13 = (7, 6)$, $14 = (7, 7)$	audit support, not a classification theorem
negative control	named subsystem-reflection rows have full centered rank $62 = 27 + 35$	separator belongs to the parabolic subgroup row, not just the reflection set

Thus the E_7 row contributes one compact control-plus-separator lesson: source-exact pair channels do not make embedded row placement redundant. The row is not a standalone E_7 theorem, not a theorem about $\mathrm{Sp}_6(2)$, not a reflection-subgroup classification, not a root-system matroid novelty claim, not an E_8 or Type-E family theorem, not a 56-weight/Gosset/minuscule route, not a classifier, and not a tiling result.

15 A Projective E_8 Source-Exact Control, Parabolic Fusion, and Subsystem Separator

The preceding E_7 row shows that source-exact pair channels do not make embedded parabolic placement redundant. The E_8 row changes the lesson again. On the 120 projective root pairs, the low pair/flat channels are already source-exact, and same-type standard simple-parabolic placements fuse under the projective Weyl action. Nevertheless, source-aware subsystem rows still separate placement once non-parabolic rows are admitted as source rows.

Let Π_{120} be the 120 antipodal pairs in the E_8 root system. The source package constructs the projective Weyl action, the projective root graph Γ_{E_8} from the squared Gram channel, the rank-two flat-size pair channel, the decomposition of $\mathbb{Q}^{\Pi_{120}}$, and selected standard-parabolic and coordinate-D_8 subsystem rows. Classical Coxeter and reflection-subgroup background is standard; see [BB05, DL11]. Root-system matroid automorphism context is prior art, not a claim of this exhibit; see [SFT11].

Proposition 15.1 (source-exact channels, standard-parabolic fusion, and subsystem-row separation in the projective E_8 shadow)**.** *For the source-locked projective E_8 shadow Π_{120}, the exact source replay certifies the following bounded facts.*

(a) *The carrier has 240 roots and 120 projective root pairs. The projective Weyl image has order*

$$|W(E_8)_{\mathrm{proj}}| = 348364800.$$

(b) *The projective root graph Γ_{E_8}, with edges given by squared Gram value 1, has strongly regular parameters*

$$(120, 56, 28, 24),$$

and source decomposition

$$\mathbb{Q}^{\Pi_{120}} = \mathbb{Q}\mathbf{1} \oplus U_{35} \oplus U_{84}.$$

The rank-two flat-size pair channel is equivalent to the same Gram-square pair channel, and both channels have full automorphism group $W(E_8)_{\mathrm{proj}}$.

(c) *The reflection row satisfies*

$$N_{\mathrm{Ref}} = 64I + A,$$

where A is the adjacency matrix of Γ_{E_8}; on the centered part this gives rank $119 = 35 + 84$.

(d) *The model coordinate/half split has sizes 56 and 64, but it is not invariant under the projective Weyl action. It is therefore a negative control, not an admitted point-color channel.*

(e) *Among the 255 nonempty standard simple-parabolic rows from the locked E_8 diagram, same-abstract-type and same-order multi-placement families do not separate at the rank, split, orbit-profile, or canonical weighted quotient-profile level. The structural reason is fusion: all 27 multi-placement standard-parabolic type/order groups are conjugate under $W(E_8)_{\mathrm{proj}}$.*

(f) *Source-aware subsystem rows still separate placement. The replay compares standard E_8 subdiagram rows with subdiagram rows inside a coordinate D_8 subsystem and certifies five same-type, same-order parabolic-versus-D_8 witnesses:*

$$4A_1,\ A_3 + 2A_1,\ 2A_3,\ A_5 + A_1,\ A_7.$$

The cleanest witness is the A_7 pair

$$\mathrm{Par}_{1234567} \quad \textit{and} \quad \mathrm{D8}_{1234567}.$$

Both have abstract type A_7 and order 40320*, but their fingerprints on* Π_{120} *differ:*

$$[8, 28, 28, 56], \qquad \text{rank} = 3 = (1, 2), \qquad \perp\text{-complement } 0,$$

versus

$$[1, 28, 28, 28, 35], \qquad \text{rank} = 4 = (1, 3), \qquad \perp\text{-complement } A_1,$$

where the ordered pairs record the (U_{35}, U_{84}) *ranks.*

Source-locked proof sketch. The exact replay constructs the E_8 roots in the scaled integer model, quotients by antipodal pairs, computes Γ_{E_8}, checks its strongly regular parameters and eigenspace dimensions, and verifies that the rank-two flat-size pair channel is equivalent to the Gram-square pair channel. The projective Weyl source action is generated by simple reflections, and the external Sage replay verifies that the full graph/channel automorphism group has the same order. The reflection-row identity is then evaluated on the two nontrivial source summands. For standard parabolics, the replay computes all 255 quotient-rank fingerprints and then certifies by a projective Weyl orbit computation that every same-type, same-order multi-placement family fuses. Finally, the coordinate-D_8 subsystem-row scout recomputes the standard parabolic rows against E_8 data and compares them with D_8-internal subdiagram rows, producing the displayed A_7 witness and the four support witness types. The D_8 coordinate/half point color remains rejected separately. □

Thus the E_8 row contributes a compact source-row lesson: when the low channel is already source-exact and standard parabolic placements fuse, the grammar can still see source placement through explicitly admitted subsystem rows. The row is not a standalone E_8 theorem, not a Type-E family theorem, not a full reflection-subgroup or root-subsystem classification, not a new Dynkin–Borel–de Siebenthal–Oshima result, not a root-system matroid novelty claim, not a classifier, and not a tiling result.

16 A/B, Dihedral, and Source-Layer Grammar Contrast

The A/B/D and reflection-shadow material enters this paper as grammar contrast. Its purpose is to show how the same admission discipline reacts when the source changes. The signed-shadow routing table and signed-balance notes are the source locks for the Type-B/C contrast. The dihedral vertex-shadow row adds a rank-two uncolored reflection-shadow contrast. The length-colored G_2 row adds a source-channel contrast: the finite set is still concrete, but the short/long partition is part of the data. The projective F_4 row adds a source-layer contrast on one finite set: the geometric projective Weyl row and the source matroid automorphism row have different centered behavior. The projective H_3 row adds a non-crystallographic source-channel contrast: flat-size incidence and Gram-angle geometry give different automorphism rows on the same 15 projective root pairs. The projective H_4 row adds the selected 60-pair analogue: flat-size and Gram-square channels have different full color automorphism groups on the same projective 600-cell root shadow. The projective D_5 row adds a control in the opposite direction: pair-level data over-automorphizes the 20-point projective root shadow, while A_2 3-flat incidence recovers the source action exactly. The projective E_6 row adds a second simply laced control: pair, flat, and circuit channels collapse to the classical source action, but the source summands U_{20} and U_{15} still separate parabolic row information hidden by total standard rank. The projective E_7 row adds the next control: pair and flat channels are already source-exact on Π_{63}, but embedded A_5 parabolic rows of the same abstract type and order have different U_{27}/U_{35} fingerprints. The projective E_8 row closes the local E-series comparison: pair and flat channels are already source-exact on Π_{120}, standard parabolic placements of the same type fuse, and non-parabolic subsystem rows still separate source placement.

Table 14: Evidence carried by the projective E_8 source-exact control, parabolic fusion, and subsystem separator.

channel or witness	certified finite fact	grammar reading
carrier	240 roots, 120 projective root pairs	largest simply laced projective root-shadow row used here
projective source row	$\lvert W(E_8)_{\mathrm{proj}}\rvert = 348364800$	target source action
pair/flat channels	Γ_{E_8} has parameters $(120, 56, 28, 24)$; Gram-square and rank-two flat-size channels have automorphism group 348364800	low pair/flat data is already source-exact
source split	$\mathbb{Q}^{\Pi_{120}} = \mathbb{Q}\mathbf{1} \oplus U_{35} \oplus U_{84}$	row data is read on two nontrivial source summands
reflection row	$N_{\mathrm{Ref}} = 64I + A$, rank $119 = 35 + 84$	source-exact carrier still has selected row identities
standard parabolics	255 rows; 27 multi-placement type/order groups; all fuse under $W(E_8)_{\mathrm{proj}}$	no same-type standard-parabolic separator is possible there
main subsystem witness	$\mathrm{Par}_{1234567}$ and $\mathrm{D8}_{1234567}$: both A_7, both order 40320, orbit profiles $[8, 28, 28, 56]$ and $[1, 28, 28, 28, 35]$, ranks $3 = (1, 2)$ and $4 = (1, 3)$	same abstract subsystem type/order does not determine the source-aware finite-shadow fingerprint
support witnesses	$4A_1$, $A_3 + 2A_1$, $2A_3$, $A_5 + A_1$, and A_7 parabolic-versus-D_8 pairs	bounded witness family, not a classification theorem
negative color control	coordinate/half point split 56/64 is not invariant	subsystem rows are admitted source rows; the point color is not

The contrast is deliberately asymmetric. The Type-A row has a published theorem source, the Type-B/C row has a closed bounded artifact plus the self-contained elementary proof of the signed mirror-union row in Proposition 6.1, the dihedral row has elementary cyclic/reflection algebra in Proposition 7.1, the G_2 row has the source-channel facts in Proposition 8.1, the projective F_4 row has the source-layer facts in Proposition 9.1, the projective H_3 row has the flat-size/Gram-channel facts in Proposition 10.1, the projective H_4 row has the flat-size/Gram-square channel facts in Proposition 11.1, the projective D_5 row has the pair-channel/A_2-incidence control facts in Proposition 12.1, the projective E_6 row has the channel-collapse/source-split facts in Proposition 13.1, the projective E_7 row has the source-exact/embedded-parabolic facts in Proposition 14.1, and the projective E_8 row has the source-exact/fusion/subsystem-separator facts in Proposition 15.1. The D-side signed-reversal material belongs to the public companion theorem route owned by P15[Bab26c]. This paper uses it only as a pointer and does not import theorem statements or proofs from it.

Table 15: How the same grammar test behaves across the current source rows, I.

source	centered channel	visible behavior	boundary
Type-A row	one standard channel V_{std}	balance and Specht $(n-1,1)$ vanishing coincide; odd side is source-bounded	no all-odd theorem and no 5568 bridge
modular centered row	augmentation, reduced-quotient, and lattice-qualified Smith channels	carrier-coprime Smith torsion can collapse finite-field rank while rational rank stays nonzero	no all-layer Type-A, Latin/Sudoku/M19, modular Specht, classifier, or tiling theorem
signed Type-B/C row	ambient standard splits into pair-standard and reflection channels	mirror-union balances for every $n \geq 3$ by the elementary X15-002 proof	six numerical rows remain bounded; no D-side theorem import
dihedral row	one vertex standard channel for $I_2(m) \leq S_m$	rational rank and SNF^L detect mirror-angle arithmetic	no general Coxeter or Weyl theorem
length-colored G_2 row	global channel plus short/long orbit-centered channels	the length line blocks global balance, while blockwise centering recovers the dihedral transfer	no projective quotient or uncolored dodecagon replacement

Table 16: How the same grammar test behaves across the current source rows, II.

source	centered channel	visible behavior	boundary
projective F_4 row	global, length-centered, and source matroid layers on Π_{24}	G_{proj} has one length residual; G_{mat} cancels it by a source length-swap coset	no F_4, 24-cell, flat-row, or Weyl classification
projective H_3 row	flat-size and Gram-angle channels on Π_{15}	flat-size symmetry has order 120, Gram symmetry has order 60, while the standard block is blind	no H_3, non-crystallographic Coxeter, matroid, classifier, or tiling theorem
projective H_4 row	flat-size and Gram-square channels on E_{60}	flat-size symmetry has order 14400, Gram-square symmetry has order 7200, and a pair witness lies in one flat-size orbit but two Gram-square orbits	no H_4, 600-cell, Weyl, matroid-automorphism discovery, classifier, or tiling theorem
projective D_5 row	pair channel and A_2 3-flat incidence on Π_{20}	pair symmetry has order 122880, while A_2 incidence recovers $W(D_5)_{\text{proj}}$ of order 1920	no standalone D_5, Type-E, $E_6/E_7/E_8$, classifier, or tiling theorem

In this sense the grammar has teeth: it can admit bounded finite shadows while refusing to smuggle in neighboring theorem routes whose source locks are not yet closed. It also shows real changes of behavior. The Type-A row balances through one centered channel and is parity-fragile on the odd side; the signed row must balance two channels at once, but the mirror-union does so for every $n \geq 3$; the dihedral row is elementary, but it exposes rational-rank versus lattice-qualified Smith data; the G_2 row shows that source partitions can be mathematically active rather than decorative; the projective F_4 row shows that the admitted source layer can change the centered-rank interpretation on the same finite set; the projective H_3 row shows that the ordinary standard block can be blind even when two source channels define different finite rows; the projective H_4 row shows that the same carrier of 60 projective points can have a 14400-

Table 17: How the same grammar test behaves across the current source rows, III.

source	centered channel	visible behavior	boundary
projective E_6 row	pair, flat, circuit, and source-split channels on Π_{36}	pair/flat/circuit channels all have source action order 51840, while U_{20}/U_{15} splits separate rows of equal size and total rank	no E_6, Schlaefli, 27-line, root-matroid novelty, classifier, or tiling theorem
projective E_7 row	pair/flat channels and embedded parabolic rows on Π_{63}	pair/flat channels have source action order 1451520, while two A_5 rows of order 720 have ranks $5 = (3, 2)$ and $6 = (3, 3)$	no E_7, $\mathrm{Sp}_6(2)$, reflection-subgroup, root-matroid, E_8, classifier, or tiling theorem
projective E_8 row	pair/flat channels, standard parabolic rows, and coordinate-D_8 subsystem rows on Π_{120}	pair/flat channels have source action order 348364800; standard parabolic same-type placements fuse; an A_7 parabolic row and a coordinate-D_8 A_7 row have ranks $3 = (1, 2)$ and $4 = (1, 3)$	no E_8, Type-E, root-subsystem classification, classifier, or tiling theorem

element flat-size source symmetry but only a 7200-element Gram-square geometric symmetry; the projective D_5 row shows that pair-level data may be too coarse even in a simply laced shadow, while A_2 incidence restores the source row; the projective E_6 row shows that channel collapse to the classical source action does not make source-aware summand data redundant; the projective E_7 row shows that source-exact pair channels still do not determine embedded parabolic row fingerprints; and the projective E_8 row shows that when standard parabolic placements fuse, source-aware subsystem rows can still carry a finite-shadow separator.

17 Value and Limits of Φ

The value of Φ is operational. In the source-locked audits, it records finite structure not visible from coarser summaries alone. The Type-A, modular centered, signed Type-B/C, dihedral, length-colored G_2, projective F_4, projective H_3, projective H_4, projective D_5, projective E_6, projective E_7, projective E_8, and centered-operator artifacts all provide finite evidence that this channel is useful for comparison.

Table 18: What the worked witnesses show, I.

source	finite observation	allowed conclusion
Type-A Fourier witness	same class histogram; ranks 2 and 1 at $(3, 1)$	Φ is not determined by class histograms
modular centered witness	$I_7(1, 1)$, $p = 2$, $\mathrm{SNF}^L = [4, 4, 8]$, rank $3 \to 0$	rational centered rank does not determine modular rank
signed Type-B/C witness	same signed-cycle histogram; ranks 4 and 2 at $(5, 1)$	the same value test survives in a signed finite shadow
signed mixed layers	$BI_3(0, 1)$ has rank mass 300, below the size-B_3 hash-ordered control 715	signed incidence rows carry visible finite structure
dihedral $m = 8$ witness	same elementwise dihedral reflection-class histogram	centered standard ranks 5 and 3 detect relative mirror separation
dihedral $m = 9$ witness	same rational centered standard rank 8	SNF^L separates 1^8 from $1^6, 2^2$
length-colored G_2 witness	short-axis and long-axis rows are ambient-conjugate after forgetting length	source profiles distinguish fixed short roots from fixed long roots
projective F_4 witness	G_{proj} and G_{mat} act on the same 24 points but have centered ranks 1 and 0	the admitted source layer matters before rank data is interpreted

17.1 A controlled X05 exhibit: M19 and `lat565`

The X05 Sudoku/Latin row gives a separate controlled example of the same point. For an order-nine square L, take $X = S_9$ acting by symbol relabeling and let

$$E_{\gamma,L}(i, j) = \gamma(L(i, j)) - 5$$

Table 19: What the worked witnesses show, II.

source	finite observation	allowed conclusion
projective H_3 witness	flat-size and Gram-angle channels act on the same 15 points with automorphism orders 120 and 60, while $(14,1)$ has rank 0	the source channel can matter even when the ordinary standard block is blind
projective H_4 witness	flat-size and Gram-square channels act on the same 60 projective root pairs with automorphism orders 14400 and 7200	source-channel data can separate matroid/incidence symmetry from geometric Gram symmetry
projective D_5 witness	pair channel has automorphism order 122880, while A_2 incidence preservers have order 1920 and equal $W(D_5)_{\mathrm{proj}}$	source recovery may require triple/flat incidence, not just pair-level data
projective E_6 witness	Par_{012} and Par_{0135} have size 24 and total rank 8, but U_{20}/U_{15} splits $(6,2)$ and $(5,3)$	source-aware summand data can see parabolic row information hidden by total standard rank
projective E_7 witness	Par_{12345} and Par_{12347} are both A_5 rows of order 720, but have ranks $5=(3,2)$ and $6=(3,3)$ on U_{27}/U_{35}	same abstract parabolic type and order need not determine the embedded finite-shadow fingerprint
projective E_8 witness	$\mathrm{Par}_{1234567}$ and $\mathrm{D8}_{1234567}$ are both A_7 rows of order 40320, but have orbit profiles $[8,28,28,56]$ and $[1,28,28,28,35]$, with ranks $3=(1,2)$ and $4=(1,3)$ on U_{35}/U_{84}	same abstract subsystem type and order need not determine the source-aware finite-shadow fingerprint
M19/X05 exhibit	blind rank-locus mining repeatedly recovers the same order-72 group G_0 when the full coset occurs	Φ can recover a hidden group from centered-operator data
`lat565` X05 exception	local $32+8+22$ centered-certificate stratification	Φ separates a hidden-coset block from non-coset centered rank-drop certificates

be the centered operator, restricted to the standard subspace. The fingerprint package records the rank-drop locus

$$\Gamma_{\mathrm{rank}}(L) = \{\gamma \in S_9 : \operatorname{rank} E_{\gamma,L} < 8\}$$

together with the hidden stabilizer recovered by blind coset mining. This is the same grammar slot Φ, now applied to a centered-operator sibling of the Type-A rows.

The published Sudoku operator paper isolates the original M19 rank-locus phenomenon; see [Bab26j]. The centered Latin-square paper supplies the sibling determinant/rank-obstruction context for the Latin side of the centered-operator row[Bab26b]. The FCIG census source then asks whether M19 is isolated. Its durable, source-locked answer is that the large order-72 hidden coset is not a one-off. Whenever that full coset occurs in the corrected census, the hidden group is the same fixed subgroup $G_0 \leq S_9$, the affine symmetry group of the natural $1,\ldots,9$ value grid. In the corrected 1000+1000 census, this full G_0-coset appears in both Sudoku and Latin samples. The visible magic-band refinement is observed only on the Sudoku side. Complement relabeling is the baseline symmetry of the centered rank-drop condition.

The companion M19 census package now records the exceptional Latin square `lat565` as a local finite-certificate stratification

$$\Gamma_{\mathrm{rank}}(\texttt{lat565}) = C_{32} \sqcup C_8 \sqcup R_{22}, \qquad 62 = 32 + 8 + 22.$$

Here C_{32} is an order-32 paired-row hidden coset outside G_0, C_8 is a signed four-row parallelogram block, and R_{22} consists of eleven centered nullity-one complement-pair certificates. Only C_{32} is group/coset-structured; the latter pieces are local centered certificates, not a second hidden group or a family theorem.

This exhibit is useful because Φ recovers a recurring finite group from the data without being handed the group in advance, and because `lat565` shows the same channel separating a hidden-coset block from non-coset centered certificates inside one object. It is not used here as a theorem pillar. The frequencies are empirical, the mechanism selecting the arrays is open, and the broader out-of-G_0 population remains outside this paper.

Table 20: M19/X05 census and `lat565` facts used here.

item	recorded status
full hidden coset	order 72; recurring signal, not a single M19 accident
recovered group	same fixed $G_0 \leq S_9$ whenever the full coset occurs
corrected census	1000 Sudoku samples and 1000 Latin samples inspected in the source package
magic-band refinement	observed on the Sudoku side only
`lat565` exception	local $32 + 8 + 22$ stratification; only the 32-point block is a hidden coset; the 8-point and 22-point pieces are centered linear certificates only

17.2 Source quotients require coupling data

The quotient mechanism is classical. Higman fixes the arbitrary commutative coefficient ring in Section 6 and gives the unnormalised block-row-sum quotient in Section 8, Eq. (8.1)[Hig70, §6, p. 11; §8, Eq. (8.1), p. 20]; Godsil records the equitable-partition identity $AH = HB$, the invariant cell-constant subspace, and spectral divisibility[God10, §5.1, Eq. (5.1.1) and Lem. 5.1.2]; and Haemers supplies the classical quotient/interlacing context[Hae95, §2]. P13 uses this mechanism only as an interface guardrail showing which source and coupling data must be retained; it makes no novelty claim for quotient existence, spectral inheritance, or interlacing.

Several rows above compare a row with a quotient row. A block partition $\Omega = \bigsqcup_B \Omega_B$ compatible with a centered operator N (constant block-column sums) induces a quotient operator N_π on the blocks, and, after choosing one base point $b(B) \in \Omega_B$ per block, the anchored augmentation lattice splits into a kernel lattice and a lifted quotient lattice, over which N becomes block triangular,

$$N \sim \begin{pmatrix} N_K & \beta \\ 0 & N_\pi \end{pmatrix},$$

where N_K is the restriction to the kernel lattice and N_π here denotes the quotient operator on its own anchored lattice. The grammar question is when the two endpoint fingerprints determine the total one, and the honest answer is a guardrail: only together with the coupling β.

The underlying descent bookkeeping can be stated without normalized averaging. Let $q : \mathbb{Z}[\Omega] \to \mathbb{Z}[\pi]$ send every point to its block and write A_X for the augmentation lattice of a finite set X. Then

$$0 \longrightarrow K_\pi := \bigoplus_{B\in\pi} A_{\Omega_B} \longrightarrow A_\Omega \xrightarrow{q} A_\pi \longrightarrow 0$$

is split exact after base points are chosen, and an operator descends exactly when

$$qN = N_\pi q \iff N(\ker q) \subseteq \ker q.$$

These are the classical quotient/equitable conditions in augmentation language. For a section $s : A_\pi \to A_\Omega$, put

$$\beta_s = N_A s - s N_{\pi A}.$$

If $s_h = s + ih$, where $i : K_\pi \hookrightarrow A_\Omega$, direct block multiplication gives

$$\beta_{s_h} = \beta_s + N_K h - h N_{\pi A}.$$

Consequently, for $v \in \ker N_{\pi A}$, the connecting class

$$\delta(v) = [\beta_s(v)] \in \operatorname{coker} N_K$$

is independent of the chosen section. This class measures one obstruction, but it does not classify integral equivariant complements.

The modular-centered row $I_7(1,1)$ supplies the useful falsifier. In its integral split augmentation basis,

$$N_K = 0_3, \qquad N_A = \begin{pmatrix} 0 & X \\ 0 & 2X \end{pmatrix}, \qquad X = \begin{pmatrix} 0 & -4 & -4 \\ -4 & 0 & -4 \\ -4 & -4 & 0 \end{pmatrix}.$$

Here $\delta_{\mathbb{Z}} = 0$, but the section equation $\beta_s + N_K h - hN_{\pi A} = 0$ has only $h = \frac{1}{2} I_3$ over $\mathbb{Q}$ and no integral solution. Thus a zero connecting map does not force an integral equivariant complement. Independently, elementary left–right unimodular operations reduce N_A to $X \oplus 0_3$; the determinantal divisors $4, 16, 128$ recover the nonzero anchored Smith factors $[4, 4, 8]$ from the 3×3 block, without a full ambient Smith reduction.

Proposition 17.1 (source-side splitting test)**.** *For each block B write*

$$d(B) = Ne_{b(B)} - \sum_{B'} (N_\pi)_{B'B}\, e_{b(B')},$$

a vector supported in the kernel lattice and computable from the base-point columns of N and the quotient matrix alone, with no Smith computation. If all the $d(B)$ coincide, then $\beta = 0$ for the base-point section, N is conjugate over $\mathbb{Z}$ to $N_K \oplus N_\pi$ on the anchored lattice by a unitriangular change of section, and $\mathrm{SNF}^L(N)$ is the Smith form of the direct sum: the multiset of elementary divisors is the union of the endpoint multisets, with the invariant factors re-chained by gcd/lcm.

Proof. On the lifted basis vector $h_C = e_{b(C)} - e_{b(C_0)}$ a direct computation gives $\beta(h_C) = d(C) - d(C_0)$, so equal defects force $\beta = 0$; the splitting statement is then block-diagonal bookkeeping, and the Smith form of a direct sum is classical. ☐

The hypothesis is not removable as a general source-side test, and endpoint data alone never suffices. The three-point row $N = \begin{pmatrix} 3 & 1 & 0 \\ -1 & 1 & 0 \\ -2 & -2 & 0 \end{pmatrix}$ with blocks $\{1, 2\}$ and $\{3\}$ has both anchored endpoint operators equal to (2), yet its anchored total operator is $\left(\begin{smallmatrix} 2 & 1 \\ 0 & 2 \end{smallmatrix}\right)$ with $\mathrm{SNF}^L = [1, 4]$: the total cokernel is $\mathbb{Z}/4$, while the uncoupled direct sum gives $[2, 2]$, that is $\mathbb{Z}/2 \oplus \mathbb{Z}/2$. The missing datum is precisely the extension class of the coupling.

For completeness, the residue description used here has an elementary proof. After endpoint Smith transforms write the block matrix as

$$\begin{pmatrix} A & C \\ 0 & B \end{pmatrix}, \qquad A = \mathrm{diag}(a_i), \quad B = \mathrm{diag}(b_j).$$

Endpoint-preserving block-unitriangular row and column operations replace C by $C + YB + AZ$. Entrywise, this replaces c_{ij} by $c_{ij} + y_{ij} b_j + a_i z_{ij}$, so the invariant is

$$[c_{ij}] \in \mathbb{Z}/(a_i, b_j) \;\cong\; \mathbb{Z}/\gcd(a_i, b_j)\mathbb{Z}.$$

Conversely, Bézout's identity shows entrywise that two couplings with the same residues differ by such a pair Y, Z. Thus the residue matrix is exactly the coupling class relative to the chosen endpoint Smith bases, not a new structure theorem.

Deciding splitting in general is Sylvester-equation solvability [Rot52, Gus79]; when the endpoint determinants are coprime the coupling never matters [New74, Thm. 2]; and the possible middle types at a shared prime form the classical Green–Klein/Littlewood–Richardson set [Kle68]; see also [Sch07, Thms. 4.1–4.2]. This subsection is therefore an interface guardrail, not a new descent or transfer theory: no shared-prime law and no novelty are claimed.

The same audits also fix the limit. The fingerprint is not a complete invariant, rational rank is not the whole fingerprint, and rank data is not used as a decision oracle for unrelated structural questions. This section records the value and the guardrails together.

18 Boundaries and Future Routes

The grammar leaves several routes outside the current claim boundary. The signed-reversal program is now a public theorem route owned by P15 and remains pointer-only here. P18 and P17 are future companion pointers, not admitted X11/X12 rows. The exceptional S_6 routes are closed in the bounded P07 and P33 companion packages but their theorems are not imported. Higher reflection-shadow routes beyond the present exhibits and the blocked numerical bridge remain future work unless their own source locks, proof boundaries, and replay packages close.

The downstream P28 length-coloring theorem, P29 golden-ring refinement, P30 exceptional census, and P32 E_7 placement study are likewise companion routes[Bab26h, Bab26g, Bab26e, Bab26d]. They refine existing source rows or expose limitations of their descriptive fingerprints; none creates a new X-ID or transfers its theorem into P13.

Table 21: Future routes and open debts kept outside the present claim boundary.

route or debt	current role	admission condition before promotion
signed-reversal route	public companion theorem route	P15 owns the theorem; kept pointer-only here
P18 nonconvex route	future companion pointer	remains `assigned_p13_label=null`; a separate P13 admission gate would be required
P17 motion/contact route	future companion pointer	finite CL3 candidate does not itself create an admitted P13 row
higher reflection-shadow routes beyond the bounded exhibits	future grammar branches	need source locks and non-overlap with this paper's dihedral, G_2, projective F_4, projective H_3, projective H_4, projective D_5, projective E_6, projective E_7, and projective E_8 exhibit claims
M19 and `lat565`	controlled X05 exhibit	`lat565` is local $32 + 8 + 22$ only; broader out-of-G_0 mechanisms remain outside this paper
shifted-diagonal row	balance-control exhibit	all-n statement remains a proof obligation
5568 bridge candidate	blocked numerical coincidence/bridge candidate	requires an identified morphism or explicit non-bridge closeout
exceptional S_6 benchmark	public companion certificate routes	P07/P33 own their bounded claims; no theorem import into P13
source-quotient coupling route	closed guardrail exhibit (Proposition 17.1)	any transfer-law promotion needs genuinely non-split source families and its own admission gate; the mechanisms are classical

The modular centered row is a compact channel exhibit only. It records that rational centered rank is not enough once finite-field reduction is admitted: the grammar must name the characteristic, the augmentation channel, the reduced quotient when p divides the carrier size, and the lattice-qualified Smith form. It does not claim an all-layer Type-A modular theorem, a Latin/Sudoku/M19 mechanism, a modular Specht novelty theorem, a classifier, or a tiling result.

The X05/M19 exhibit has its own boundary. This paper may use it to show that Φ can recover a recurring hidden group in a centered-operator rank locus and can record, for `lat565` only, a finite $32 + 8 + 22$ centered-certificate stratification. This paper does not claim a general Sudoku or Latin theorem, a classification of X05 rank drops or out-of-G_0 objects, a second hidden group for the full `lat565` rank locus, or any tiling consequence.

The dihedral row also has its own boundary. It is a rank-two vertex-shadow exhibit inside S_m, not a general Coxeter or Weyl theory. The special case $I_2(6)$ is not used as a theorem about G_2, and the lattice-qualified Smith data in Tables 6 and 18 is not promoted to a classifier.

The length-colored G_2 row is likewise bounded. It uses the twelve-root finite shadow only with its source partition $\Omega_{\mathrm{s}} \sqcup \Omega_{\mathrm{l}}$. The projective quotient and the uncolored dodecagon are controls, not replacement sources, because both forget the short/long channel. This paper does not claim a general Weyl-group grammar, a projective theorem, or any tiling consequence from the G_2 exhibit.

The projective F_4 row is bounded in the same way. It uses only the 24-point projective root shadow Π_{24}, the long/short source partition, the projective Weyl layer G_{proj}, and the source matroid automorphism layer G_{mat}. It does not classify F_4 rows, block/triality rows, flat-incidence rows, 24-cell geometry, Weyl groups, or tilers, and it does not claim the F_4 matroid automorphism group as a new discovery.

The projective H_3 row is bounded in parallel. It uses only the 15-point projective root shadow Π_{15}, the flat-size edge-coloring channel, the Gram-angle edge-coloring channel, and the ordinary S_{15} Specht scout as descriptive fingerprint data. It does not prove a general H_3, non-crystallographic Coxeter, root-system matroid, classifier, or tiling theorem. The public source posture is the bundled finite coordinate/source-channel replay over $\mathbb{Q}(\phi)$; no external A–O notation is used as a page-locked proof source here.

The projective H_4 row is even more tightly bounded. It uses only the 60 antipodal pairs in the H_4 root system, viewed as a projective 600-cell root shadow, and only the two pair-color channels recorded in the selected source package: rank-two flat size and squared Gram value. It does not claim a new computation of $\mathrm{Aut}(M(H_4))$, a full H_4 theory, a 600-cell classification, an E-type or icosian theorem, a Weyl/Coxeter theorem, a classifier, or a tiling result. The public posture is the bundled de-labelled H4 replay/package, not a general H4 engine.

The projective D_5 row is a control exhibit only. It uses the 20 antipodal projective pairs in the D_5 root system, the pair-level Gram/flat-size channel, the coordinate D_4 negative-control family, and the 40 projective A_2 3-flat incidences. It does not claim a standalone D_5 theorem, a general Type-E theorem, an $E_6/E_7/E_8$ transfer, a matroid automorphism classification, a classifier, or a tiling result. The A_2 incidence is a triple/flat-incidence channel, not another pair channel.

The projective E_6 row is also bounded. It uses only the 36 antipodal projective pairs in the E_6 root system, the projective root graph, the flat-size and circuit channels, and the source summands U_{20} and U_{15}. It does not claim a standalone E_6 theorem, a Type-E family theorem, a Schlaefli/double-six/27-line/cubic-surface route, a new root-system matroid result, a classifier, or a tiling consequence. The parabolic witness is explicitly not a same-type or same-orbit indistinguishability witness. The projective E_7 row is bounded in the same way. It uses only the 63 antipodal projective pairs in the E_7 root system, the Gram-square/projective root graph, the rank-two flat-size pair channel, the source summands U_{27} and U_{35}, and selected embedded simple-parabolic rows. It does not claim a standalone E_7 theorem, a theorem about $\mathrm{Sp}_6(2)$, a reflection-subgroup classification, a root-system matroid novelty result, an E_8 or Type-E family theorem, a 56-weight/Gosset/minuscule route, a classifier, or a tiling consequence. The parabolic separator is explicitly an embedded-row fingerprint witness, not a classification of parabolics. The projective E_8 row is bounded likewise. It uses only the 120 antipodal projective pairs in the E_8 root system, the Gram-square/projective root graph, the rank-two flat-size pair channel, the source summands U_{35} and U_{84}, the standard simple-parabolic fusion replay, and selected coordinate-D_8 subsystem rows. It does not claim a standalone E_8 theorem, a Type-E family theorem, a full reflection-subgroup or root-subsystem classification, a new Dynkin–Borel–de Siebenthal–Oshima result, a root-system matroid novelty result, a classifier, or a tiling consequence. The subsystem separator is explicitly a bounded source-row fingerprint witness, not a classification of E_8 subsystems. The shifted-diagonal material and the 5568 bridge candidate remain fenced. The shifted-diagonal row is useful as a balance-control exhibit, but the all-n statement is still a proof obligation. The numerical bridge involving 5568 is not used as a theorem or as evidence of an identified morphism.

This boundary is part of the result of the interface. The grammar is useful only if it admits source-locked finite rows while refusing rows whose finite shadow, representation channel, or fingerprint package is not yet defined. If later drafting removes the worked examples in Tables 1 to 15, 18, 20 and 21, the honest fallback is a shorter interface note rather than a full paper.

A Source and Instance Manifest

This appendix records the finite-shadow instance matrix used by the paper. Public scholarly rows are cited in the bibliography. Companion artifact rows are included only as bounded source rows for the interface; they are not cited as independent mathematical authority unless separately released.

The third column below gives a descriptive registry role and, where relevant, pointer posture. It is not the canonical admission type. Canonical admission is typed separately as `admitted` or `not_admitted`, with an independent `pointer_status`, in `artifacts/source_instance_manifest.json`.

Table A.1. Finite-shadow instance matrix.

ID	row	registry role / pointer posture	role and boundary
X01	Type-A poset cones $L(P)$	core/prose	Type-A chamber context; no new extension-set theorem.
X02	Type-A extension tilers	prose/exhibit	Exact-cover context; no Φ-tiling test.
X03	subgroup/coset rows	core exhibit	Controls for rank mass and left-ideal background.
X04	finite S_4 witness	core exhibit	Published bounded witness only.
X05	Sudoku/M19 rank loci	core/prose	Controlled M19 exhibit; `lat565` locally source-locked as $32 + 8 + 22$; no X05 or out-of-G_0 classification.
X06	Latin determinant-defect loci	prose	Centered-operator sibling; no proof import.
X07	magic/endpoint defect loci	future	No unresolved finite-field theorem promoted.
X08	one-incidence layers $I_n(k, l)$	core	Type-A theorem boundary; no 5568 bridge.
X09	prefix/shadow subsets	prose	Folded into finite-witness provenance.

ID	row	registry role / pointer posture	role and boundary
X10	exceptional S_6 benchmark	companion pointer	P07 owns the fixed P038 result and P33 owns the finite $57 = 19 + 38$ certificate; P13 imports neither theorem nor proof and makes no all-n tiling claim.
X11	nonconvex Coxeter-pure tilers	future companion pointer	P18 is a future companion and remains unassigned to P13; `assigned_p13_label=null`.
X12	motion/contact atlas objects	future companion pointer	P17 has a finite CL3 candidate, but no P13 admission or theorem import.
X13	Fourier/Specht fingerprints	core artifact	Descriptive Φ, lattice-qualified SNF^L, ranks, and rank mass; exact Type-A values are pinned by the static X13 certificate.
X14	grammar interface	prose/meta	Organizes rows; not a theorem by itself.
X15	signed Type-B/C shadows $B_n \leq S_{2n}$	core artifact	Six bounded numerical rows plus the separately proved mirror-union balance proposition; no broader Coxeter claim or P15 theorem import.
X16	modular centered operators	core exhibit	Characteristic-sensitive augmentation/reduced-quotient/SNF^L row; $I_7(1,1)$ carrier-coprime torsion collapse; no all-layer Type-A, Latin/Sudoku/M19, modular Specht, classifier, or tiling claim.
X17	signed reversal noncentral layer	public companion pointer	P15 owns the published theorem route; no proof or theorem is imported here.
X18	dihedral reflection shadows $I_2(m) \leq S_m$	core exhibit	Vertex-shadow row; balance, two-mirror rank, SNF^L, and common-precedence closure; no general Weyl claim.

ID	row	registry role / pointer posture	role and boundary
X19	length-colored G_2 root shadow	core exhibit	Source-partition row; short/long channel, orbit-centered balance, two-mirror transfer, and source-profile witness; no general Weyl or projective claim.
X20	24-point projective F_4 root shadow	core exhibit	Source-layer row; G_{proj} length residual, G_{mat} length-swap cancellation, block-kernel and flat-incidence supporting evidence; no F_4, 24-cell, Weyl, or classifier claim.
X21	15-point projective H_3 root shadow	core exhibit	Source-channel row; flat-size symmetry 120, Gram-angle symmetry 60, standard block blind, first higher Specht separator $(13, 2)$; no H_3, non-crystallographic Coxeter, matroid, or classifier claim.
X22	60-pair projective H_4 root shadow	core exhibit	Source-channel row; flat-size channel symmetry 14400, Gram-square channel symmetry 7200, and a pair witness in one flat-size orbit but distinct Gram-square orbits; no H_4, 600-cell, Weyl, matroid-automorphism discovery, classifier, or tiling claim.
X23	20-point projective D_5 root shadow	core exhibit	Control row; pair-channel symmetry $122880 = C_2^{10} : S_5$, projective $W(D_5)$ order 1920, and 40 A_2 3-flat incidences recovering the source action; no standalone D_5, Type-E, $E_6/E_7/E_8$, classifier, or tiling claim.
X24	36-point projective E_6 root shadow	core exhibit	Control row; pair/flat/circuit channel collapse to source action order 51840, while U_{20}/U_{15} split detects parabolic row information hidden by total standard rank; no E_6, Schlaefli, 27-line, root-matroid novelty, classifier, or tiling claim.
X25	63-point projective E_7 root shadow	core exhibit	Control-plus-separator row; Gram-square and rank-two flat-size pair channels have source action order 1451520, while two embedded A_5 parabolic rows of order 720 have ranks $5 = (3, 2)$ and $6 = (3, 3)$; no E_7, $\mathrm{Sp}_6(2)$, reflection-subgroup, root-matroid, E_8, classifier, or tiling claim.
X26	120-point projective E_8 root shadow	core exhibit	Source-exact/fusion/subsystem row; Gram-square and rank-two flat-size pair channels have source action order 348364800, standard parabolic placements fuse, while an A_7 parabolic row and a coordinate-D_8 A_7 row have orbit profiles $[8, 28, 28, 56]$ and $[1, 28, 28, 28, 35]$; no E_8, Type-E, root-subsystem classification, classifier, or tiling claim.

Public and companion pointers. The public Type-A standard-space paper, the public S_4 finite witness paper, the public Type-A poset-cone/extension-tiler paper, and the public Latin-square and Sudoku operator papers are scholarly pointers for rows X01, X02, X04, X05, X06, X08, and related prose rows. The label P038 is overloaded across the wider source archive: the public Type-A poset-cone paper contains a proved extension-tiler witness with that label, while X10 is served by the bounded P07/P33 companion routes; those results retain their companion ownership and are not imported as P13 theorems. The Fourier/Specht, modular centered, signed Type-B/C, dihedral vertex-shadow, length-colored G_2, projective F_4, projective H_3, projective H_4, projective D_5, projective E_6, projective E_7, and projective E_8 rows are companion source packages for the present grammar paper. The M19 census package is a companion source-lock pointer for the X05 exhibit and is not used as public mathematical authority unless separately exported.

B Reproducibility Notes

This repository contains the LaTeX source, compiled title-named PDF, public boundary notes, source manifest, public replay artifacts, and SHA-256 manifest for the paper snapshot. The paper is an interface paper: its finite numerical rows are replayable in the cited public companion repositories or in bounded companion artifacts summarized by Section A. No theorem in this paper depends on treating those companion artifacts as public mathematical authority.

The paper can be rebuilt from `paper/main.tex` with the title-named job target `A_Finite-Shadow_Grammar_for_Finite_Centered_Incidence_Geometry`. Use `pdflatex` with that `-jobname` value, then `bibtex` on the same job name, followed by three additional `pdflatex` passes with the same `-jobname` option. The expected output is

`paper/A_Finite-Shadow_Grammar_for_Finite_Centered_Incidence_Geometry.pdf`.

Data and Code Availability

The companion repository contains the LaTeX source, compiled PDF, source manifest, claim-boundary notes, and SHA-256 manifest for this paper snapshot. The finite computations cited here are either in the cited public companion repositories or recorded as source-locked companion source packages summarized by the source manifest; this repository does not claim a standalone replay engine for every upstream row.

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